

Two-phase multiscale numerical framework for modeling thin films on curved solid surfaces in porous media

Zhipeng Qin ^{a,b,*}, Soheil Esmaeilzadeh ^a, Amir Riaz ^c, Hamdi A. Tchelepi ^a

^a Department of Energy Resources Engineering, Stanford University, Stanford, CA 94305, USA

^b Department of Geophysics, Stanford University, Stanford, CA 94305, USA

^c Department of Mechanical Engineering, University of Maryland, College Park, MD 20742, USA

ARTICLE INFO

Article history:

Received 11 February 2020

Received in revised form 31 March 2020

Accepted 8 April 2020

Available online 17 April 2020

Keywords:

Multiphase flow

Porous media

Immersed boundary method

Level-set method

Thin films

Curved solid surfaces

ABSTRACT

Complex solid geometries, subgrid thin films, and interfacial deformations are critical for studying multiphase flows in porous media. However, it is challenging to accurately predict the flow evolution in porous media because of the difficulties in capturing the fluid-solid and fluid-fluid interactions in complex-shaped confinements at the pore-scale. In this work, we propose a multiscale continuum framework enabling the accurate simulation of the multiphase flow physics in complex porous media. Within this framework, we couple an efficient and accurate incompressible Navier-Stokes solver developed on a fixed Cartesian grid with a level-set method designed for capturing the interface between immiscible two-phase flows at low Capillary numbers. To capture the effects of complex solid boundaries on the flow dynamics, we employ an immersed boundary method based on a direct forcing approach. Most importantly, we develop a subgrid-scale thin-film model on a cylindrical coordinate around the curved solid surfaces to resolve the thin liquid films below the grid resolution and capture their effects on the fluid-fluid interfaces. The proposed multiscale framework illustrates how the embedded subroutines such as the level-set approach, the immersed boundary method, and the thin-film model can be coupled in order to capture the crucial physics involved in multiphase flow within porous media at different length scales.

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1. Introduction

Immiscible fluid-fluid displacement through confined porous pathways is involved within many physical and chemical processes in industrial applications and natural phenomena where accurately understanding, capturing, and controlling the underlying involved physics is crucial. Contaminant transport [1], enhanced oil recovery [2,3], shale gas production [4], geological CO₂ sequestration and storage [5], fuel cell design [6], hydrology and water infiltration into soil [7], and microfluidics [8,9] are some prominent examples. In order to accurately capture the flow behavior in these applications, it is necessary to accurately resolve the underlying physics at the pore-scale where the interface-level dependencies dictate the larger-scale transport of immiscible multiphase flow.

In investigating the pore-scale interfacial dynamics there are a number of significant challenges involved due to the combined effects of capillary forces, wettability, and viscous forces [10], the dynamics associated with solid shape complex-

* Corresponding author at: Department of Geophysics, Stanford University, Stanford, CA 94305, USA.

E-mail addresses: zhipengq@stanford.edu (Z. Qin), soes@stanford.edu (S. Esmaeilzadeh), ariaz@umd.edu (A. Riaz), tchelepi@stanford.edu (H.A. Tchelepi).

ities [11], the molecular scale features such as thin films [12] and moving contact lines [13], and the large scale interfacial merger and break-up [14]. Such mechanisms altogether make the macroscopic displacement patterns highly dependent on the strength of the viscous forces relative to capillary forces (*i.e.*, Capillary number), the affinity of solid geometry toward different fluid phases (*i.e.*, wettability), and the structure of the pore geometry [15,16].

In order to overcome these challenges, existing studies have developed different approaches for investigating the pore-scale interfacial evolution. The three main approaches that have been widely used are (i) particle based models, *e.g.*, lattice Boltzmann method (LBM) [17–19] and smooth particle hydrodynamics (SPH) [20–22], (ii) topology-based models, *e.g.*, pore network approach [23,24], and (iii) continuum models, *e.g.*, direct pore-scale modeling [25–27]. As reviewed by Joekar-Niasar and Hassanizadeh [24], the continuum model leads to more accurate interface calculations with respect to the other two models. More importantly, the continuum models can address the problem of spurious currents which poses a significant challenge for the numerical simulations at low Capillary numbers.

Among all the factors affecting the interfacial dynamics in porous materials, the solid geometry configuration has a significant impact [28–31]. Porous media, although often considered to be a fractal type structure [32], inherently is built by the coexistence of different pore structures in a disordered manner. The solid geometry configuration in porous media controls the amount of storage capacity within aquifers and subsurface systems [33,34]. Moreover, the connectivity of the pore structures controls the fluid flow and transport through confined formations [35]. Solid geometry also governs the dependency of the bulk flow properties on the pore-scale characteristics [36].

The existence of subgrid-scale thin films on solid surfaces is another critical factor affecting the interfacial dynamics and multiphase flow transport [37–42]. In oil/water/gas systems in porous media, the wetting films may contribute to different phenomena influencing the oil recovery. The presence of wetting thin films can limit the overall recovery efficiency by inducing the non-wetting phase entrapment. For instance, Constantinides and Payatakes [43] showed that the preexisting flow of thin films can lead to phase separation ahead of the main bulk flow advancement and cause capillary trapping of the non-wetting phase during imbibitions. On the other hand, the existence and rupture of the wetting thin films may lead to molecular scale wettability alteration which has recently gained significant attention in the area of enhanced oil recovery (EOR) processes [44,45]. In both of the aforementioned two areas, thin films of the order of micrometers affect the larger scale dynamics. The most challenging aspect of simulating such thin films in porous media is their extremely thin dimension of micrometer scale. This not only imposes experimental limitations and makes direct observations difficult, particularly within a three-dimensional porous medium, but also the numerical simulations become impractical requiring a very high resolution (comparable to the thin-film scales) in order to fully capture the thin-film interfacial dynamics. Hence, a significant computational effort is required to faithfully simulate multiphase flow in porous media over length and time scales of practical interest. One way of overcoming the computational costs of resolving the wide ranges of length-scales present in porous media is using multiscale approaches [46–50].

In the present work, we propose a multiscale continuum model for simulating the immiscible two-phase pore-scale flow using subgrid-scale thin films on curved solid surfaces. The proposed framework consists of three main components, namely, an immersed boundary based Navier-Stokes solver, a level-set two-phase flow approach, and a subgrid thin-film lubrication model. In the Navier-Stokes solver, using a uniform Cartesian grid, we associate the standard projection method that is used for solving the Navier-Stokes equations with a semi-implicit treatment of the viscous stress tensor [51,52]. In favor of computational efficiency, the semi-implicit treatment of the viscous stress helps to overcome the time-step restriction commonly present in the pore-scale low Reynolds flow regimes with high viscosity contrasts. Additionally, we apply the ghost fluid method [53] to capture the interfacial pressure jump across the fluid-fluid interface. In order to properly capture the flow dynamics in the presence of complex porous structures, we couple an immersed boundary method, based on a direct forcing strategy [52,54], with the Navier-Stokes solver in order to capture the pore structure. The direct forcing strategy, more efficient than other immersed boundary methods [55,56], has been used in previous pore-scale flow studies (*see, e.g.*, [57,58]). Within the immersed boundary method used in the current framework, similar to the approach proposed by Balaras [59], we impose the boundary conditions on the solid surface by a linear interpolation which is applicable to arbitrary solid shapes. Our previous studies [54] have demonstrated that this particular method of imposing solid surface boundary conditions achieves a second-order accuracy in calculation of the flow dynamics around the embedded solid geometries within the computational domain.

The present work focuses on the capillary dominated multiphase flow at the pore-scale, in which case, the Capillary number, representing the ratio of viscous forces to surface tension forces acting across the interface, could be as low as 10^{-4} [60,61]. At such low Capillary numbers, an accurate calculation of the curvature at the fluid-fluid interface is crucial for avoiding erroneously induced numerical spurious currents [62]. In the proposed model, we employ two-phase level-set method [63] in order to track the advancing fluid-fluid interface. Compared to other interface tracking strategies, such as the volume-of-fluid method [64] and the front tracking approach [65], the level-set method represents the fluid-fluid interface as a continuous function, guaranteeing higher accuracy in curvature calculations. In order to preserve mass conservation during the advection of the level-set function, we use a topology-preserving strategy, proposed by Qin et al. [66], which maintains the level-set function as a signed distance function using a projection based reinitialization strategy.

In the present work, in a multiscale manner, we numerically resolve the evolution of the subgrid thin films on curved solid surfaces by coupling a thin-film lubrication approximation with the immersed boundary based Navier-Stokes solver and the level-set two-phase flow approach. In a similar approach as proposed by Abu-Al-Saud et al. [67], we assume that a preexisting thin-film layer, which has been experimentally observed ([68,69]), covers the solid surface in order to re-

move the unbounded stress singularity that occurs when the fluid-fluid interface comes into contact with the solid surface [70]. Assuming that the solid surface is spherical, we propose and discretize a thin-film evolution equation in a cylindrical coordinate for the 2D and 2D-axisymmetric simulations. Using a linear interpolation approach, our proposed framework couples the large scale dynamics with the micro-scale subgrid thin film by transferring the parameters between Cartesian and cylindrical coordinates.

The current manuscript is ordered as follows: In Section 2, we first describe the governing equations of the proposed model including the Navier-Stokes equations, the level-set method, and the subgrid thin-film lubrication approximation. Afterwards, in Section 3 we introduce the numerical methodologies used to solve the governing equations at the discretized level. In Section 4, we evaluate the accuracy of the proposed model and carry out validation for static, advancing (imbibition) and receding (drainage) interfaces on a curved surface in both 2D and 2D-axisymmetric domains. Finally, we study the effects of several physical parameters on the thin-film evolution in Section 4.

2. Governing equations

2.1. Interfacial-advection and Navier-Stokes equations for immiscible two-phase flow

In the present work, we consider incompressible and immiscible two-phase flow under isothermal conditions and in electrokinetic equilibrium. In our two-phase flow numerical approach, the level-set function is used to capture the evolution of the fluid-fluid interfaces and the interfacial dynamics [71]. The level-set ($\phi(\mathbf{x}, t)$) is an implicit function equal to zero at the interface, conventionally negative in one fluid phase and positive in the other phase. The interfacial dynamics and the spatial and temporal evolution of the phase distribution is captured by solving the level-set advection equation as

$$\frac{\partial \phi(\mathbf{x}, t)}{\partial t} + \mathbf{v} \cdot \nabla \phi(\mathbf{x}, t) = 0, \quad (1)$$

where $\mathbf{v}$ is the flow velocity vector, $\phi(\mathbf{x}, t)$ is the level-set function, $\mathbf{x}$ is the spatial coordinates, and t is time.

Under isothermal conditions and assuming that the fluid phases are incompressible, the conservation of mass and momentum (Navier-Stokes equations) in the computational domain, Ω , are expressed as

$$\nabla \cdot \mathbf{v} = 0, \quad (2)$$

and

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{\nabla p}{\rho} + \frac{1}{\rho} \nabla \cdot \left[\mu (\nabla \mathbf{v} + \nabla \mathbf{v}^T) \right] + \mathbf{g} + \mathbf{f}_B, \quad (3)$$

where $\mathbf{v}$ represents the flow velocity field, p is pressure, ρ is the density of the fluid, μ is the dynamic viscosity of the fluid, $\mathbf{g}$ is the gravitational acceleration, and $\mathbf{f}_B$ is an external body force exerted on the fluid accounting for the presence of solid boundaries.

The fluid's density and dynamic viscosity values are computed based on the level-set function value $\phi(\mathbf{x}, t)$ as

$$\begin{aligned} \rho(\phi) &= (\rho_1 - \rho_2)H(\phi) + \rho_2, \\ \mu(\phi) &= (\mu_1 - \mu_2)H(\phi) + \mu_1, \end{aligned} \quad (4)$$

where the subscripts 1 and 2 indicate the corresponding fluid phases, and $H \in [0, 1]$ is a smooth transition function defined as

$$H(\phi) = \frac{1}{2} \left[1 - \operatorname{erf}\left(\frac{\phi}{\varepsilon}\right) \right], \quad (5)$$

where $\operatorname{erf}(\cdot) \in [-1, 1]$ is the error function, and ε is a computational grid dependent smoothing hyperparameter that represents the thickness of the smoothed interface. Here, ε is set to be 1.5 times of the grid size.

The jump condition of the normal stress at the fluid-fluid interface can be written as

$$[p]_I = \sigma \kappa + 2[\mu]_I \mathbf{n}^T \cdot \nabla \mathbf{v} \cdot \mathbf{n}, \quad (6)$$

where $[\cdot]_I$ indicates the sharp jump across the interface, κ is the interface curvature, σ is the surface tension, and $\mathbf{n}$ is the normal vector of the interface. Above interfacial jump condition (Eq. (6)) simplifies to $[p]_I = \sigma \kappa$, because we use diffused viscosity around the interface (as in Eq. (4)) [53]. The normal vector $\mathbf{n}$ and the curvature κ of the interface can be computed using the value of the level-set function $\phi(\mathbf{x}, t)$ respectively as

$$\mathbf{n} = \frac{\nabla \phi}{|\nabla \phi|}, \quad (7a)$$

and

$$\kappa = \nabla \cdot \mathbf{n} = \nabla \cdot \frac{\nabla \phi}{|\nabla \phi|}. \quad (7b)$$

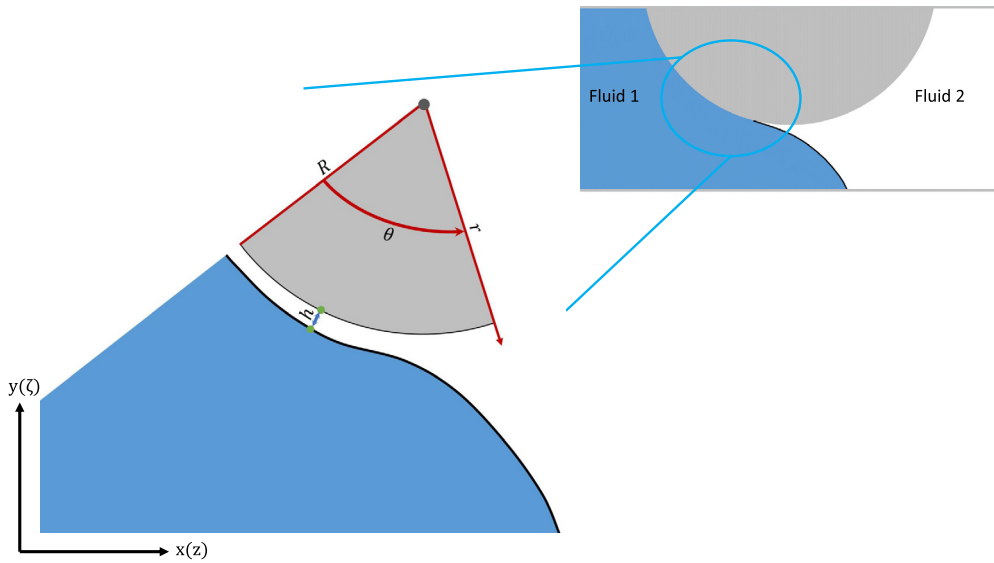


Fig. 1. Zoomed in meniscus profile around a spherically curved surface, illustrating the computation of the thin-film thickness, h , based on multiscale coupling $r - \theta$ cylindrical coordinates and $x - y$ Cartesian coordinates or $z - \zeta$ axisymmetric coordinates. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

2.2. Thin-film modeling: a lubrication approximation

In this work, we model the subgrid thin-film evolution on the solid boundaries using a lubrication approximation described by a nonlinear partial differential equation derived from the Navier-Stokes equations. Fig. 1 shows the thin-film deposited on a spherically curved solid surface in the presence of two fluid phases, and illustrates a cylindrical coordinate system used for the thin-film discretization in 2D and 2D-axisymmetric domains. In the present study, we focus on the low Reynolds number regime ($Re = \rho UL/\mu < 10$) where the viscous forces dominate the inertial forces, and consider the thickness of the thin film on curved solid boundaries to be small (e.g., several orders of magnitude smaller than the pore size). Accordingly, the velocity component in the normal direction with respect to the thin film will be negligible, and the velocity in the tangential direction can be considered parabolic. Therefore, the equation that governs the evolution of the thin-film thickness h can be derived as [72]

$$\frac{\partial h}{\partial t} + \nabla \cdot \mathbf{Q} = 0, \quad (8)$$

where h is the thin-film thickness, $\mathbf{Q}$ is the flow flux inside the thin film, and t is time. Due to the small thin-film thickness, $\mathbf{Q}$ can be computed as the product of the average flow velocity within the thin film ($\langle \mathbf{v} \rangle_{\text{avg}}$) and the thin-film thickness (h) as $\mathbf{Q} = h \langle \mathbf{v} \rangle_{\text{avg}}$.

The average flow velocity within the thin film ($\langle \mathbf{v} \rangle_{\text{avg}}$) is computed by integrating the parabolic velocity in the thin film ($\mathbf{v}$) over the thin-film thickness. Accordingly, by applying a no-slip velocity boundary condition at the solid surface, and a normal stress jump and a continuous tangential shear stress at the thin-film surface, the average flow velocity within the thin film $\langle \mathbf{v} \rangle_{\text{avg}}$ becomes

$$\langle \mathbf{v} \rangle_{\text{avg}} = \frac{1}{\mu} \left(\frac{h}{2} \boldsymbol{\tau} - \frac{h^2}{3} \nabla p \right), \quad (9)$$

where $\boldsymbol{\tau}$ represents the shear stress on the thin-film surface, p is pressure, h is thin-film thickness, and μ is the dynamic viscosity (which is the viscosity of fluid 2 in Fig. 1). The shear stress in the tangential direction (τ), by considering the tangential-to-solid velocity component to be w , is proportional to the gradient of w in the normal direction as

$$\tau = \mu \frac{\partial w}{\partial r}, \quad (10)$$

where μ is the dynamic viscosity, r is the radial component of the cylindrical coordinate system illustrated in Fig. 1. The pressure jump across the thin-film interface (i.e. the difference between the pressure within the thin film below its surface (p) and the pressure outside the thin film (p_o)) has two components, namely, capillary pressure and disjoining pressure, which can be expressed as

$$p - p_o = -\sigma\kappa + \Pi(h), \quad (11)$$

where κ is the thin-film curvature, σ is the surface tension, and $\Pi(h)$ is the disjoining pressure.

By substituting the average flow velocity within the thin film $\langle \mathbf{v} \rangle_{\text{avg}}$ (Eq. (9)) and the pressure jump across the thin film (Eq. (11)) into the evolution equation of the thin-film thickness (Eq. (8)), we obtain the following partial differential equation that governs the evolution of thin-film thickness in a cylindrical coordinate as

$$\frac{\partial h}{\partial t} = -\frac{1}{\mu R} \frac{\partial}{\partial \theta} \left\{ \frac{h^3}{3} \left[\frac{1}{R} \frac{\partial(\sigma\kappa)}{\partial \theta} - \frac{1}{R} \frac{\partial \Pi(h)}{\partial \theta} \right] + \frac{h^2 \tau}{2} \right\}, \quad (12)$$

where R is the radius of the curved solid surface and θ represents the azimuthal component of the cylindrical coordinate.

In the present work, we consider films around regularly shaped curved surfaces, e.g., circular ones in 2D case or spherical ones in 3D case. Under the assumption of film being so thin such that the ratio of the average thin-film thickness ($\bar{h}$) to the characteristic length (L) be much less than 1 (i.e., $\epsilon = \bar{h}/L \ll 1$), Howell [73] defines the curvature of the free surface as

$$\kappa \sim \kappa_1 + \kappa_2 + \frac{\epsilon L}{a} (\kappa_1^2 + \kappa_2^2) h + \frac{\epsilon a}{L} \nabla_s^2 h + O(\epsilon^2), \quad (13)$$

where κ_1 and κ_2 represent the principal curvatures and a is the typical radius of curvature. For the thin-film curvature on a straight solid boundary see ref. [67].

In a 2D case, the circular solid surface in x - y coordinate could be represented as

$$0 = -R + \sqrt{(x - x_o)^2 + (y - y_o)^2}, \quad (14)$$

where R is the radius of the surface, x_o and y_o represent the coordinates of circle center. Defining the characteristic length scale as $L = a$, and considering a cylindrical coordinate system as shown in Fig. 1, the curvature of a 2D circular surface becomes

$$\kappa = -\frac{1}{R} + \epsilon \left(\frac{1}{R^2} h + \frac{1}{R^2} \frac{\partial^2 h}{\partial \theta^2} \right). \quad (15)$$

In capillary tubes (2D-axisymmetric), the curved solid surface can be expressed in z - ζ axisymmetric coordinates (shown in Fig. 1) as

$$\varphi = -R + \sqrt{(z - z_o)^2 + (\zeta - \zeta_o)^2}, \quad (16)$$

where z_o and ζ_o are the coordinates of surface center. Accordingly, the mean curvature of the curved solid surface, H , becomes

$$2H = -\nabla \cdot \frac{\nabla \varphi}{|\nabla \varphi|} = - \left[\frac{\varphi_z^2 \varphi_{\zeta\zeta} + \varphi_{\zeta}^2 \varphi_{zz} - 2(\varphi_z \varphi_{\zeta} \varphi_{z\zeta})}{(\varphi_z^2 + \varphi_{\zeta}^2)^{3/2}} + \frac{1}{\zeta} \frac{\varphi_{\zeta}}{\sqrt{\varphi_z^2 + \varphi_{\zeta}^2}} \right]. \quad (17)$$

Considering the relationship between the mean curvature and the principal curvature, $H = (\kappa_1 + \kappa_2)/2$, we approximately define $\kappa_1 \approx \kappa_2 = H$ and set the characteristic length L as $1/H$. Accordingly, the Eq. (13) can be rewritten for the curvature of a 2D-axisymmetric surface as

$$\kappa = 2H + \epsilon \left[2H^2 h + \frac{1}{R^2} \frac{\partial^2 h}{\partial \theta^2} \right]. \quad (18)$$

Similar to the case of having thin-films on straight solid boundaries [67], in the presence of repulsive Van der Waals forces, the disjoining pressure between two parallel curved surfaces can be expressed as $\Pi(h) = A/(6\pi h^3)$, where A represents the Hamaker constant. As mentioned in [67], in a smooth capillary tube with straight wall, the disjoining pressure should be modified to an axisymmetric form, $\Pi'(h)$, by multiplying with a correction term, Λ , as

$$\Pi'(h) = \Lambda \Pi(h) = \frac{\Upsilon}{\Upsilon - h} \Pi(h), \quad (19)$$

where Υ is the radius of the capillary tube in an axisymmetric setup. Considering the fact that the thin-film thickness is extremely smaller than the radius of the tube and the curved solid surface (i.e., $h \ll \Upsilon$ and $h \ll R$), the correction term Λ would be approximately unity, $\Lambda \approx 1$. Accordingly, we will use the same disjoining pressure for both 2D and 2D-axisymmetric cases in the present work (i.e., $\Pi'(h) = \Pi(h)$).

3. Numerical methodology

In this section, we introduce details of the three components within our numerical framework, namely, the Cartesian grid solver of the Navier-Stokes equations (section 3.1), the subgrid-scale thin-film model (section 3.2), and the level-set method (section 3.3).

3.1. Cartesian grid solver of Navier-Stokes equations

In this work, the incompressible Navier-Stokes equations (Eqs. (2)–(3)) are solved based on the projection method [74] with approximate factorization [75] and fractional time-stepping [51] on a Cartesian staggered grid. In the framework of the classical three-step (n , $n + 1/2$, and $n + 1$) projection method, [74], first a provisional velocity field ($\hat{\mathbf{v}}$) at $n + 1/2$ is calculated as

$$\frac{\hat{\mathbf{v}} - \mathbf{v}^n}{\Delta t} = -(\mathbf{v}^n \cdot \nabla) \mathbf{v}^n + \frac{1}{\rho(\mathbf{x})} \nabla \cdot [\mu(\nabla \hat{\mathbf{v}} + (\nabla \mathbf{v}^n)^T)] + \mathbf{g} + \mathbf{f}_{IB}^{n+1} \quad \text{in } \Omega, \quad (20)$$

$$\hat{\mathbf{v}} = \mathbf{b}^{n+1} \quad \text{on } \Gamma, \quad (21)$$

where $\mathbf{f}_{IB}^{n+1}$ and $\mathbf{b}^{n+1}$ are the boundary condition on the solid surface, Γ , determined by the immersed boundary method that is presented in detail in Appendix A.

As elaborated in Appendix A, the viscous term in Eq. (20) is split into implicit and explicit parts. This special treatment, removes the time-step restriction associated with the viscous term. In 2D, we rewrite Eq. (20) as

$$\left[1 - \frac{\Delta t}{\rho(\mathbf{x})} \frac{\partial}{\partial x} \left(\mu \frac{\partial}{\partial x} \right) - \frac{\Delta t}{\rho(\mathbf{x})} \frac{\partial}{\partial y} \left(\mu \frac{\partial}{\partial y} \right) \right] \hat{\mathbf{v}} = \mathbf{v}^n - \Delta t \left\{ (\mathbf{v}^n \cdot \nabla) \mathbf{v}^n + \frac{1}{\rho(\mathbf{x})} \nabla \cdot [\mu(\nabla \mathbf{v}^n)^T] \right\} + \mathbf{g} + \mathbf{f}_{IB}^{n+1}. \quad (22)$$

To reduce the computation cost and memory requirements, we replace the large sparse coefficient matrix on the left hand side of Eq. (22) with tridiagonal matrices [51,52]. Accordingly, the left hand side of Eq. (22) could be written as

$$\left[1 - \frac{\Delta t}{\rho(\mathbf{x})} \frac{\partial}{\partial x} \left(\mu \frac{\partial}{\partial x} \right) \right] \left[1 - \frac{\Delta t}{\rho(\mathbf{x})} \frac{\partial}{\partial y} \left(\mu \frac{\partial}{\partial y} \right) \right] \hat{\mathbf{v}} = \mathbf{v}^n - \Delta t \left\{ (\mathbf{v}^n \cdot \nabla) \mathbf{v}^n + \frac{1}{\rho(\mathbf{x})} \nabla \cdot [\mu(\nabla \mathbf{v}^n)^T] \right\} + \mathbf{g} + \mathbf{f}_{IB}^{n+1}. \quad (23)$$

We use a second-order accurate boundary condition [51] in order to compute the provisional velocity ($\hat{\mathbf{v}}$) using the Thomas algorithm [76] as

$$\hat{\mathbf{v}} = \mathbf{b}^{n+1} + \Delta t \frac{1}{\rho(\mathbf{x})} \nabla p^n \quad (24)$$

Afterwards, the provisional velocity ($\hat{\mathbf{v}}$) is used to calculate the velocity field $\mathbf{v}^{n+1}$ (Eq. (25)) as well as the pressure p^{n+1} (Eq. (28)) at step $n + 1$. The final velocity field $\mathbf{v}^{n+1}$ can be calculated as

$$\frac{\mathbf{v}^{n+1} - \hat{\mathbf{v}}}{\Delta t} = -\frac{1}{\rho(\mathbf{x})} \nabla p^{n+1}, \quad (25)$$

subject to

$$\nabla \cdot \mathbf{v}^{n+1} = 0 \quad \text{in } \Omega, \quad (26)$$

$$\mathbf{n} \cdot \mathbf{v}^{n+1} = \mathbf{n} \cdot \mathbf{b}^{n+1} \quad \text{on } \Gamma, \quad (27)$$

where $\mathbf{n}$ represents the normal vector of the solid surface Γ . The pressure p^{n+1} can be calculated by solving a Poisson equation for pressure using an asymmetric multi-frontal method [77] as

$$\nabla \cdot \left(\frac{1}{\rho(\mathbf{x})} \nabla p^{n+1} \right) = \frac{1}{\Delta t} \nabla \cdot \hat{\mathbf{v}} \quad \text{in } \Omega. \quad (28)$$

As mentioned earlier, due to the diffuse viscosity profile (Eq. (4)), the pressure jump across the fluid-fluid interface (Eq. (6)) becomes

$$[p]_I = \sigma \kappa, \quad (29)$$

which is captured by the ghost fluid method proposed by Kang et al. [53].

By introducing Eqs. (21) and (27) into Eq. (25), the Neumann boundary condition for the pressure on the solid surface could be obtained as

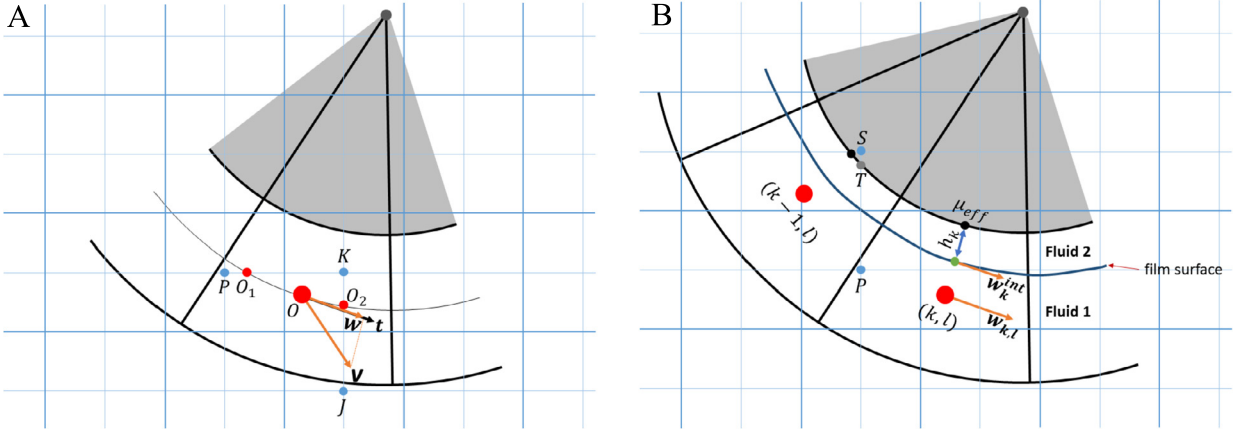


Fig. 2. (A) illustrates the translation of scalar field from Cartesian coordinate (blue lines) to cylindrical coordinate (black curves). (B) illustrates the treatment of the viscosity boundary condition. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

$$\nabla p^{n+1} \cdot \mathbf{n} = 0 \quad \text{on } \Gamma. \quad (30)$$

In the present work, we do not explicitly apply pressure boundary conditions to the curved solid surface Γ . Using the immersed boundary method described in Appendix A, we linearize all velocity components in the vicinity of Γ , where the momentum equation normal to Γ hence reduces to $dp/d\mathbf{n} = 0$.

3.2. Subgrid-scale thin-film lubrication model

In this work we consider scenarios where the solid surface is covered by a preexisting thin-film layer which prevents the interface from coming in direct contact with the solid surface, as shown in Fig. 1. For grid resolutions finer than the thin-film thickness the lubrication approximation is of course not needed because the thin film is explicitly fully resolved by direct numerical simulation. However, resolving the thin film directly is computationally expensive and prohibitive, especially for the low-capillary-number flows where the thin-film thickness can be several orders of magnitude smaller than the average length scale of the pore space. Therefore, using the lubrication approximation is an efficient strategy to capture the evolution of the thin-films at the subgrid-scales.

As illustrated in Section 3.1, the Navier-Stokes equations are solved on a uniform Cartesian grid. To solve the thin-film evolution equation built in a cylindrical coordinate system and couple the dynamics of the subgrid thin film with the flow dynamics, we need to transfer the flow parameters such as level-set value, velocity field, and fluid properties such as viscosity and density, between the Cartesian and cylindrical coordinates. In the following, in Section 3.2.1, we introduce how the flow parameters can get transferred between the two coordinate systems. Afterwards, in Section 3.2.2, we present the discretization of the subgrid thin-film evolution equation.

3.2.1. The translation between Cartesian and cylindrical coordinates

Using the structured Marker and Cell (MAC) method, we define the scalar fields, including p , ρ , μ , and ϕ , at the grid-cell centers, such as points P , Q and R in Fig. 2A. For instance, in order to transfer the level-set value ϕ from the Cartesian coordinate to the cylindrical coordinate, we first interpolate ϕ from the Cartesian grid-cell centers (i.e., points P , K and J) to the intersection points between the grid lines of the two coordinate systems (i.e., points O_1 and O_2) as

$$\begin{aligned} \Phi_{O_1} &= \frac{|O_1 K|}{|PK|} \phi_P + \frac{|PO_1|}{|PK|} \phi_K, \\ \Phi_{O_2} &= \frac{|O_2 J|}{|KJ|} \phi_K + \frac{|KO_2|}{|KJ|} \phi_J, \end{aligned} \quad (31)$$

where the subscripts represent the points where the level-sets are defined, and we use Φ instead of ϕ to represent the level-set in the cylindrical coordinate system. Afterwards, we compute the level-set at the Cylindrical grid-cell center, O , as

$$\Phi_O = \frac{\widehat{O O_2}}{\widehat{O_1 O_2}} \Phi_{O_1} + \frac{\widehat{O_1 O}}{\widehat{O_1 O_2}} \Phi_{O_2}, \quad (32)$$

where $\widehat{O O_2}$, $\widehat{O_1 O_2}$, $\widehat{O_1 O}$, $\widehat{O O_2}$ are the corresponding arc lengths between points O , O_1 , and O_2 in the cylindrical coordinate. The other scalar fields such as p , ρ , and μ can get translated from the Cartesian coordinate system to the cylindrical coordinate system in a similar manner.

Eq. (10) indicates that the θ -component (or the tangential component with respect to the solid surface) of the velocity in the cylindrical coordinate (w) is needed for the computation of the shear stress at the thin-film surface. In the structured Marker and Cell (MAC) method, the velocity components are defined at the grid-cell faces in the Cartesian coordinate system. Thus, at first, we average the velocity components at the Cartesian grid-cell centers and then interpolate them to the cylindrical coordinate grid-cell centers using the same strategy presented by Eqs. (31)–(32). Afterwards, as illustrated in Fig. 2A, we compute w by projecting the velocity at the cylindrical coordinate grid-cell centers ($\mathbf{v}$) onto the tangential direction $\mathbf{t}$ as

$$w = \mathbf{v} \cdot \mathbf{t}. \quad (33)$$

This translation of the flow parameters between the Cartesian and cylindrical coordinates is a two-way process. This means that, we also translate the necessary flow variables back from the cylindrical coordinate system to the Cartesian coordinate system upon solving the thin-film lubrication approximation (i.e., Eq. (12)) at each time-step. According to the description of our immersed boundary method (e.g., Eq. (A.3) in Appendix A), we need a boundary condition for the viscosity on the solid side nearby the solid surface, e.g., the viscosity μ_S in the center of a Cartesian grid cell S as shown in Fig. 2B. Following a similar strategy for translating flow properties between the cylindrical and Cartesian coordinates, we compute μ_S by transferring the viscosity from the cylindrical grid-cell face to the Cartesian grid-cell center as

$$\mu_S = \frac{|TS|}{|PT|} |\mu_P - \mu_T| + \mu_T, \quad (34)$$

where μ_T represents the viscosity on the cylindrical grid-cell face.

Assuming the thin film to be approximately parallel to the solid surface, we compute the viscosity on the cylindrical cell face by applying a shear-stress continuity condition to the fluids on the either sides of the thin-film surface as

$$\mu_1 \frac{\partial w}{\partial r} = \mu_2 \frac{\partial w}{\partial r}, \quad (35)$$

where r represents the radial direction in the cylindrical coordinate. As illustrated in Fig. 2B, the discretization of Eq. (33) leads to

$$\mu_1 \frac{w_{k,1} - w_k^{int}}{\Delta r/2 - h_k} = \mu_2 \frac{w_k^{int} - 0}{h_k}, \quad (36)$$

where w^{int} represents the tangential component (in θ direction) of the velocity at the thin-film surface, Δr is the grid size in the radial direction, and h is the thin-film thickness. Rearranging Eq. (36) leads to the following equation as

$$\frac{\mu_1 \mu_2 \Delta r/2}{\mu_1 h_k + \mu_2 (\Delta r/2 - h_k)} \frac{w_{k,1}}{\Delta r/2} = \mu_{eff} \frac{w_{k,1}}{\Delta r/2} = \mu_2 \frac{w_k^{int}}{h_k}, \quad (37)$$

where on the middle point of each grid-cell face in the cylindrical coordinate (represented by the black filled circles in Fig. 2B) we can obtain the following expression for μ_{eff} as

$$\mu_{eff} = \frac{\mu_1 \mu_2 \Delta r/2}{\mu_1 h_k + \mu_2 (\Delta r/2 - h_k)}. \quad (38)$$

Then we use the same strategy as Eq. (32) to interpolate the viscosity from the middle points of the grid-cell faces to the intersection points between the solid surface and the Cartesian grid lines, such as point T .

3.2.2. The discretization of the thin-film evolution equation

In the cylindrical coordinate system as shown in Fig. 3, we discretize the thin-film lubrication equation (Eq. (12)) on the grid cells shown in red, where the thin-film thickness is less than $\Delta r/2$. The level-set functions of the grid cells in the cylindrical coordinate are interpolated from the Cartesian coordinate system. In Fig. 3, the interface lies between grids $(k, 1)$ and $(k+1, 1)$. In the clockwise direction from grid cell $(k+1, 1)$, the thin-film height dips below the grid resolution, where a thin-film subgrid model is needed in order to capture the interfacial evolution of the thin-film.

The right hand side of Eq. (12) has three force components, namely, capillary, Van der Waals, and shear stress. Using a central finite-difference, the first term on the right hand side of Eq. (12) (i.e., the capillary term) could be discretized as

$$\frac{1}{R} \frac{\partial}{\partial \theta} \left[\left(\frac{h^3}{3} \right) \left(\frac{1}{R} \frac{\partial (\sigma \kappa)}{\partial \theta} \right) \right] = \frac{\sigma}{R^2} \left[\left(\frac{h^3}{3} \right)_{k+1/2} \frac{\kappa_{k+1} - \kappa_k}{\Delta \theta^2} - \left(\frac{h^3}{3} \right)_{k-1/2} \frac{\kappa_k - \kappa_{k-1}}{\Delta \theta^2} \right], \quad (39)$$

where κ is the curvature of solid surface, which is given by Eq. (15) for the 2D case and by Eq. (18) for the 3D case. Eq. (15) for the 2D case can be discretized as

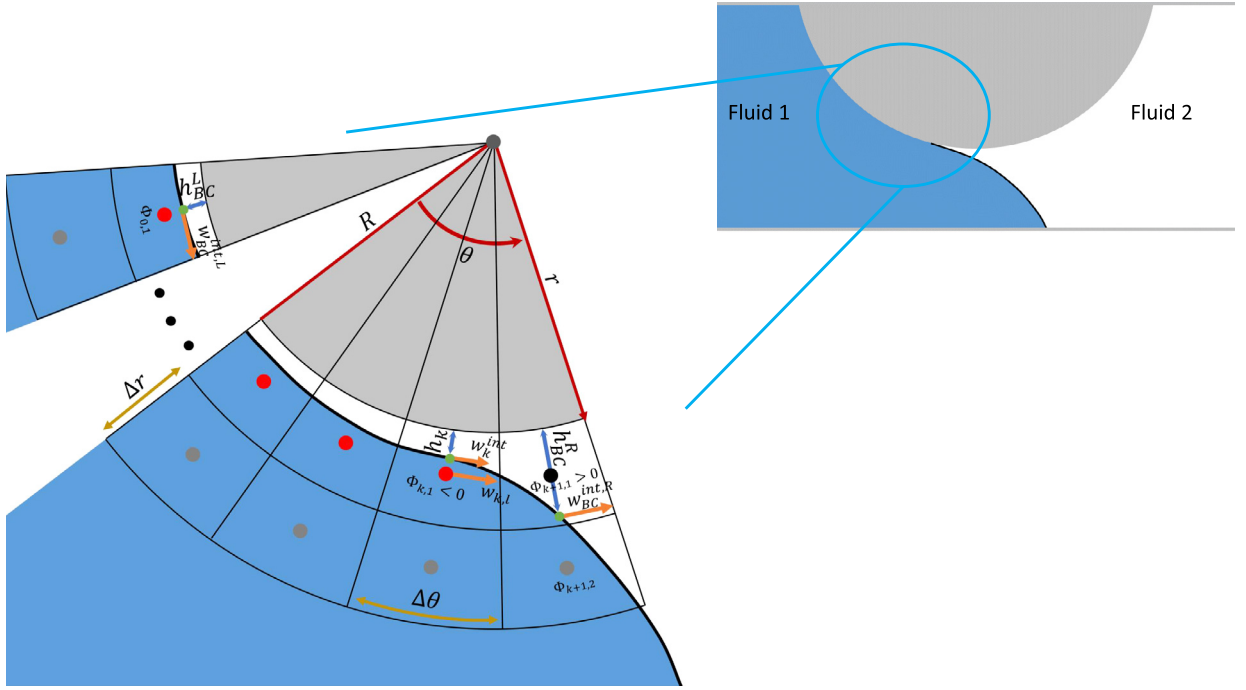


Fig. 3. Zoomed in meniscus profile around a curved solid surface in the cylindrical coordinate system. Here we illustrate the components for computation of the thin-film thickness based on the cylindrical coordinate system. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

$$\kappa_k = -\frac{1}{R} + \frac{1}{R^2}h_k + \frac{1}{R^2} \frac{h_{k+1} - 2h_k + h_{k-1}}{\Delta\theta^2}. \quad (40)$$

Similarly, Eq. (18) for the 3D case can be discretized as

$$\kappa_k = -2H_k + 2(H_k)^2h_k + \frac{1}{R^2} \frac{h_{k+1} - 2h_k + h_{k-1}}{\Delta\theta^2}, \quad (41)$$

where H_k represents the half of the surface mean curvature at θ_k , which can be found analytically using Eq. (17). In Eq. (39), the term $h^3/3$ could be referred to as the film's pseudo-diffusivity term at the grid-cell face $(1, k + 1/2)$. Using the same strategy given by Diez et al. [78] and Abu-Al-Saud et al. [67], we compute the thin-film diffusivity term at the grid-cell face $(1, k + 1/2)$ as

$$\left(\frac{h^3}{3}\right)_{k+1/2} = \frac{2}{3} \frac{h_k^2 h_{k+1}^2}{h_k + h_{k+1}}. \quad (42)$$

The second term in the right hand side of Eq. (12), the Van der Waals force, can be discretized as

$$\frac{1}{R} \frac{\partial}{\partial \theta} \left[\left(\frac{h^3}{3}\right) \left(\frac{1}{R} \frac{\partial(\Pi(h))}{\partial \theta}\right) \right] = \frac{A}{6R^2\pi} \left[\left(\frac{h^3}{3}\right)_{k+1/2} \frac{h_{k+1}^{-3} - h_k^{-3}}{\Delta\theta^2} - \left(\frac{h^3}{3}\right)_{k-1/2} \frac{h_k^{-3} - h_{k-1}^{-3}}{\Delta\theta^2} \right], \quad (43)$$

where the thin-film diffusivity term $h^3/3$ at the grid-cell face $(1, k + 1/2)$ could as well be computed using Eq. (42).

Considering the definition of shear stress given by Eq. (10) the third term in the right hand side of Eq. (12) can be discretized as

$$\frac{1}{R} \frac{\partial}{\partial \theta} \left[\frac{h^2 \tau_1}{2} \right] = \frac{1}{2R} \frac{w_{k+1}^{int} h_{k+1} - w_{k-1}^{int} h_{k-1}}{2\Delta\theta}. \quad (44)$$

To compute the velocity in θ -direction at the thin-film surface, w_k^{int} , we apply the shear-stress continuity condition given by Eqs. (35)–(36), hence we obtain

$$w_k^{int} = \frac{\mu_1 h_k w_{k,1}}{\mu_1 h_k + \mu_2 (\Delta r/2 - h_k)}. \quad (45)$$

Finally, the boundary conditions of κ , h , and w^{int} have to be specified at the two ends of the thin film where the thin-film thickness gets above the grid resolution. In the case shown in Fig. 3, we use the superscript R to indicate the boundary of thin film on the right side (i.e., located at the grid-cell $(k+1, 1)$), whereas we use the superscript L to indicate the boundary of thin-film on the left side (located at the grid cell $(0, 1)$). We employ the strategy proposed by Abu-Al-Saud et al. [67] for treating the boundary condition of the curvature. On the right side (i.e., grid-cell $(k+1, 1)$), we calculate κ_{BC}^R by averaging the level-set curvature computed from Eqs. (7) and the thin-film curvature calculated from Eq. (40) (or Eq. (41)). On the left side of thin film (i.e., grid-cell $(0, 1)$), we apply a Neumann boundary condition for the curvature as $\frac{\partial \kappa}{\partial \theta}|_{BC}^L = 0$, which allows the thin-film to move freely at the edge of the solid surface.

On the right side of thin film (i.e., grid-cell $(k+1, 1)$), thin-film thickness on the right side h_{BC}^R could be expressed as

$$h_{BC}^R = \Delta r \left(\frac{1}{2} + \frac{|\Phi_{k+1,1}|}{|\Phi_{k+1,1}| + |\Phi_{k+1,2}|} \right), \quad (46)$$

where Φ represents the level-set values in the Cylindrical coordinate system that is interpolated from the level-set values in the Cartesian coordinate system ϕ . The second term in Eq. (46) represents the subcell distance from the black point to the green point in Fig. 3. Similar to the curvature, the left boundary condition of the thin-film thickness h_{BC}^L is also imposed as a Neumann condition as $\frac{\partial h}{\partial \theta}|_{BC}^L = 0$ by which the boundary restriction on the evolution of h at the edge of the solid surface will be removed.

For the boundary condition of w^{int} on the right side of the thin film, we apply a shear-stress continuity between the grid-cells $(k+1, 1)$ and $(k+1, 2)$ as

$$\mu_1 \frac{w_{k+1,2} - w_{BC}^{int,R}}{3\Delta r/2 - h_{BC}^R} = \mu_2 \frac{w_{BC}^{int,R} - w_{k+1,1}}{h_{BC}^R - \Delta r/2}, \quad (47)$$

where by some rearrangements, $w_{BC}^{int,R}$ could be expressed as

$$w_{BC}^{int,R} = \frac{\mu_1(h_{BC}^R - \Delta r/2)w_{k+1,2} + \mu_2(3\Delta r/2 - h_{BC}^R)w_{k+1,1}}{\mu_1(h_{BC}^R - \Delta r/2) + \mu_2(3\Delta r/2 - h_{BC}^R)}. \quad (48)$$

For the left boundary of the thin film, we specify $w_{BC}^{int,L} = w_1^{int}$ for mass conservation.

3.3. Coupling the level-set method with thin-film lubrication model

After the Navier-Stokes equations are solved and the thin-film profile is resolved, the fluid-fluid interfacial evolution gets captured by the advection of the level-set function. As the present work focuses on multiphase flow at low Capillary numbers where the surface tension force significantly affects the flow dynamics, an accurate curvature computation hence is of great importance. In the present work, using the level-set approach leads to a more accurate curvature computation compared to other methods (e.g., the Volume-of-fluid method and the front-tracking method) [71,79]. The level-set method in this work has two main components: the level-set advection equation (Eq. (1)) and a reinitialization strategy.

In the present work, we use a fifth-order WENO scheme (Weighted Essentially Non-Oscillatory), proposed by Jiang and Shu [80] to solve Eq. (1) in the Cartesian coordinate system. Because there is no fluid in the solid domain, we only solve Eq. (1) on the grid-cells only in the fluid domain instead of the whole domain. However, the solution of Eq. (1) on the grid cells near the solid surface, e.g., the point “1” in Fig. 4, needs the level-set values in the solid domain for level-set derivative discretization. As illustrated in Fig. 4, we couple the level-set function with the thin-film profile in order to extrapolate the level-set values in the fluid domain onto the solid domain in the normal direction with respect to the solid surface.

On the solid grid-cells where the normal vector does not intersect with the thin film's surface, e.g., the solid grid-cell S_1 , we extrapolate the level-set value linearly as

$$\phi_{S_1} = \frac{|P_2 S_1|}{|P_1 P_2|} \phi_{P_1} + \frac{|P_1 S_1|}{|P_1 P_2|} \phi_{P_2}, \quad (49)$$

where ϕ_{S_1} , ϕ_{P_1} , and ϕ_{P_2} represent the value of the level-set function at the points S_1 , P_1 , and P_2 respectively. As the points P_1 and P_2 are not necessarily the grid-cell centers, we interpolate ϕ_{P_1} and ϕ_{P_2} from the nearby grid-cell centers, e.g., we interpolate ϕ_{P_1} from the level-set values at cell centers “1” to “4”. In the proposed model, we decide the locations of P_1 and P_2 by guaranteeing all their nearby cell centers are in the fluid domain, and define $|S_1 P_1| = |P_1 P_2|$.

On the solid grid-cells where the normal vector intersects with the thin film's surface, e.g., the solid grid-cell S_2 , we compute the level-set value as

$$\phi_{S_2} = h_O + |OS_2|, \quad (50)$$

where O represents the intersection between the normal vector and the solid surface, and h_O is the thin-film thickness normal to point O , which is interpolated from the thin-film thicknesses h_k and h_{k-1} , corresponding to the two adjacent

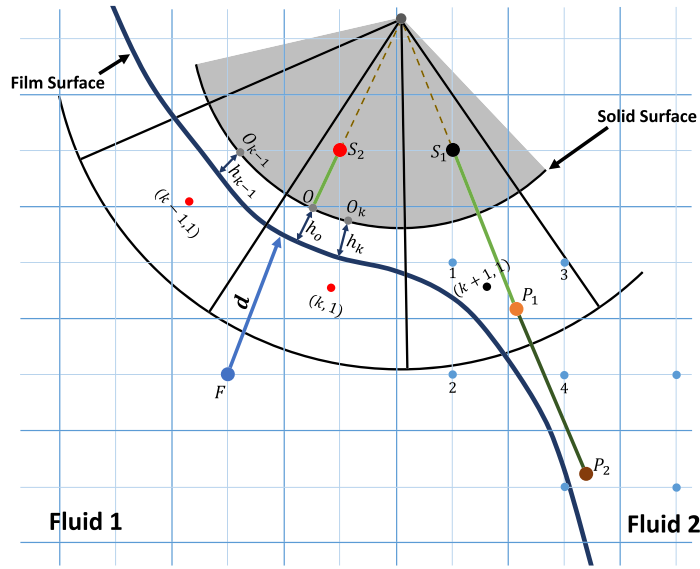


Fig. 4. Schematic description of our level-set method. We define the boundary conditions for the advection of the level-set function by extrapolating the level-set value from the fluid domain to the solid domain, e.g., from points P_1 and P_2 to point S_1 or from the corresponding thin film to point S_2 . For mass conservation, we employ a topological preserved reinitialization strategy to keep the level-set function in the fluid domain, e.g., at grid-cell F , as a signed distance function. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

cylindrical grid-cells. As we have conventionally defined the level-set function to be positive within the fluid phase 2, it would be obvious that Eq. (50) leads to a positive ϕ_{S_2} value.

The second component of the level-set approach in this work is the reinitialization step. In order to conserve mass and also maintain high accuracy in representing the fluid-fluid interface by the level-set function, we reinitialize the level-set function obtained by solving the advection Eq. (1) at each time-step. Previous studies have proposed a number of reinitialization strategies for the level-set function [71,81–83]. In the present work, we employ the topology-preserving reinitialization approach proposed by Qin et al. [66], which defines the level-set function at the Cartesian grid-cells in the fluid domain, e.g., point F in Fig. 4, as the signed distance to the fluid-fluid interface by

$$\phi_F = -\mathbf{n} \cdot \mathbf{d}, \quad (51)$$

where $\mathbf{n}$ represents the unit vector normal to the interface (or thin film's surface), and $\mathbf{d}$ is the distance vector from the grid-cell F to the interface. Considering the boundary conditions on the solid surface (i.e., Eqs. (49)–(50)), our level-set reinitialization approach maintains the level-set function as a signed distance function, which guarantees the accurate computation of the curvature.

3.4. Major steps of the proposed numerical model

Here we summarize the steps and the coupling between the Navier-Stokes solver, the subgrid thin-film lubrication approximation, and the level-set approach presented earlier. The multiscale numerical framework presented in this work consists of the following major steps:

1. The computations start by initializing the flow properties such as $\mathbf{v}^0$, p^0 , ϕ^0 , ρ^0 , and μ^0 . At each time-step n , we obtain the values of the velocity vector ($\mathbf{v}^{n+1}$) and the pressure field (p^{n+1}) at the next time step $n+1$ by solving the Navier-Stokes equations following the strategy described in Section 3.1. In this step, the curvature, κ^n , calculated based on the level-set function, ϕ^n , is used to compute the jump condition across the fluid-fluid interface (from Eq. (29)). ϕ^n is also used to define the viscosity and density of the multiphase flow (from Eq. (4)). At the Cartesian grid-cells overlapped with the subgrid thin film, we use the thin-film thickness, h^n , to compute the boundary condition of viscosity on the solid surface (Eqs. (34)–(38)) which is necessary for the discretization of the viscous term in the Navier-Stokes equations.
2. Based on $\mathbf{v}^{n+1}$ solved in the previous step, the velocity tangential to the thin film, w^{n+1} , is computed from Eq. (33). Then the thin-film profile is updated by the solution of Eq. (12) using the discretization strategy introduced in Section 3.2.2. Afterwards, the curvature (κ_k) at the grid cells in the cylindrical coordinate which include the thin film, is computed from the subgrid model (Eq. (40) for 2D cases and Eq. (41) for 2D-axisymmetric case).
3. Afterwards, the fluid-fluid interface is advected by the level-set method described in Section 3.3. In this step, the level-set function, ϕ^{n+1} , is updated by the solution of Eq. (1). On the solid surface, the boundary condition of ϕ is built using the linear extrapolation (Eqs. (49)–(50)) which requires the thin-film thickness, h^{n+1} , found previously.

Table 1

Parameters of the simulation for the static meniscus on a curved surface. The properties of the wetting (fluid 2) and non-wetting (fluid 1) fluids are similar in this case ($\mu_1 = \mu_2$ and $\rho_1 = \rho_2$). R is the Radius of the solid curve, θ and $\hat{\theta}$ are the range of the angles in radians and degrees, ρ and μ are the density and viscosity of the fluids, σ is the surface tension, A is the Hamaker constant.

R (μm)	θ (rad)	$\hat{\theta}$ ($^\circ$)	ρ (kg/m^3)	μ (cP)	σ (mN/m)	A (J)
≈ 147.8	$\in [3.70, 5.72]$	$\in [-59, 59]$	1000	2.5	50	-9.5E-20

4. The level-set function (ϕ^{n+1}) on the Cartesian grid-cells in the fluid domain is reinitialized by the topology-preserving approach (Eq. (51)) described in Section 3.3.
5. Finally, the curvature κ^{n+1} of the fluid-fluid interface by Eq. (7) is computed using the level-set function ϕ^{n+1} , for the computations in the next time-step.

4. Results and discussions

In this section, we present the results obtained using the proposed multiscale model that involves solving the Navier-Stokes equations, the level-set method, the immersed boundary approach, and the thin-film lubrication model. We first study a static meniscus on a curved solid surface in both 2D and 2D-axisymmetric cases. Afterwards, we investigate the dynamics of an advancing interface for two scenarios of drainage and imbibition in a converging-diverging capillary tube. Moreover, as a validation of our proposed multiscale model, in each case, we compare the thin-film profile obtained using the lubrication approximation as part of the multiscale model with the thin-film profile numerically resolved by refined computational grid cells spanning the subgrid resolution of the thin film.

4.1. Static thin-films in the presence of a static meniscus on a curved surface

In this section, we first study a wetting fluid thin-film deposited on a curved solid surface in the presence of a non-wetting fluid static meniscus in both 2D and 2D-axisymmetric scenarios. The size of the computational domain is $[320\mu\text{m} \times 200\mu\text{m}]$ and $[320\mu\text{m} \times 100\mu\text{m}]$ in the 2D and 2D-axisymmetric scenarios respectively. Table 1 lists the relevant parameters of the numerical simulation. We solve the Navier-Stokes-level-set governing equations (Eqs. (1)-(3)) on a Cartesian grid for the 2D and 2D-axisymmetric cases with respectively $[n_x, n_y] = [80, 50]$ and $[n_z, n_\zeta] = [80, 25]$ grid points in each direction, corresponding to grid sizes of $\Delta x = \Delta y = 4 \mu\text{m}$. We refer to this grid as the *coarse* grid. In order to capture the evolution of the subgrid thin-film profile, we solve the lubrication approximation, governed by Eq. (12), in a Cylindrical coordinate with the corresponding radial and azimuthal grid sizes of $\Delta\theta$ and Δr which will be introduced later on.

At the beginning of the simulation, we set an initial spherical shape for the macroscopic fluid interface and an initial thickness of $h_0 = \Delta r/4$ for the subgrid thin film. As the pressure jump across the fluid-fluid interface is being balanced by the surface tension force and disjoining pressure, both the macroscopic and subgrid films are expected to reach the steady state as the simulation proceeds in time. Since there is no analytical solution for the augmented Young-Laplace problem in the presence of a curved solid surface, we verify the solution of the subgrid model by comparing it with the fully resolved thin film obtained by numerically solving the Navier-Stokes-level-set on a very high resolution Cartesian mesh with $[n_x, n_y] = [640, 400]$ and $[n_z, n_\zeta] = [640, 200]$ grid points respectively for the 2D and 2D-axisymmetric cases. We refer to this grid as the *fine* grid.

In a macroscopic manner, Figs. 5A and 5C show the solution of the proposed multiscale model for the steady meniscus on a curved solid surface respectively for the 2D and 2D-axisymmetric scenarios on the coarse grid. In these scenarios, we solve the thin-film lubrication model (Eq. (12)) with the cylindrical grid sizes of $\Delta\theta \approx 0.022$ rad and $\Delta r = 8 \mu\text{m}$ (note $h_0 = \Delta r/4$). On the other hand, Figs. 5B and 5D show the fully resolved steady meniscus obtained by numerically solving the Navier-Stokes-level-set on the high resolution Cartesian grids (*i.e.* fine grid) respectively for the 2D and 2D-axisymmetric scenarios.

As a subgrid-scale comparison, in the 2D and 2D-axisymmetric scenarios, Figs. 6A and 6D compare the subgrid thin-film profile captured by the multiscale model on the coarse grid (h) with the fully resolved thin-film profile on the fine grid ($\tilde{h}$). The horizontal axis in Figs. 6A and 6D represent the azimuthal angle in degrees ($\hat{\theta}$) as illustrated in Fig. 5A. As the comparison of Figs. 6A and 6D implies, the solution of the thin-film profile obtained from the thin-film lubrication approximation on the coarse grid well matches with the fully resolved subgrid thin film on the fine grid in both 2D and 2D-axisymmetric scenarios. This shows that our proposed multiscale model accurately predicts the subgrid thin film profile on the coarse grid where the film thickness is well below the grid resolution.

Figs. 6B and 6E present the error convergence study of the subgrid thin-film lubrication model respectively for the 2D and 2D-axisymmetric scenarios as a function of the coarse grid azimuthal resolution ($\Delta\theta$). Additionally, Figs. 6C and 6F present the error convergence study as a function of the coarse grid radial resolution (Δr). In the error convergence study, the difference between the subgrid thin-film thickness obtained by the lubrication approximation (h) and the fully resolved thin-film thickness on the fine grid ($\tilde{h}$) is evaluated versus the coarse grid resolution. Such a difference can be quantified as a non-dimensionalized thickness error defined as

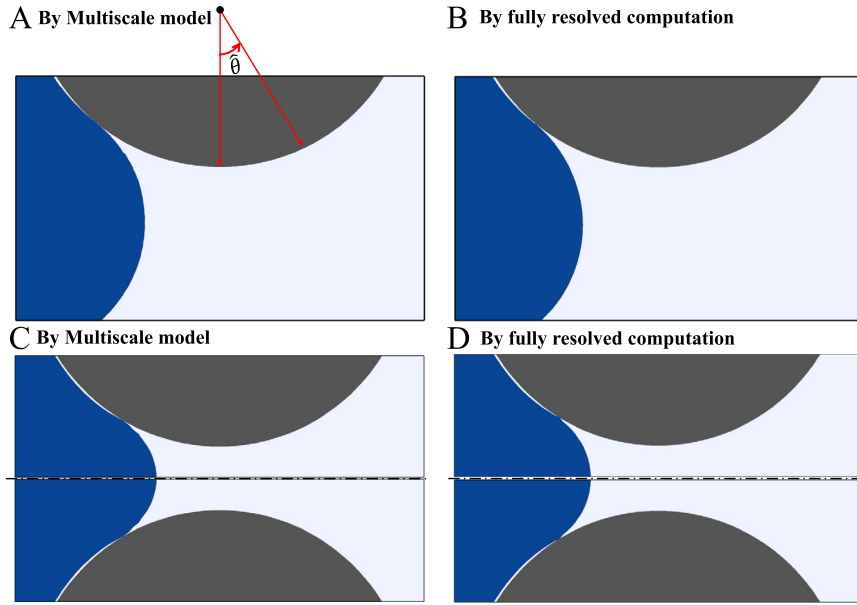


Fig. 5. The static fluid-fluid interface for an augmented Young-Laplace problem in the presence of a curved surface. A and C represent respectively in a 2D and 2D-axisymmetric cases, the subgrid thin film resolved by our multiscale model on a coarse Cartesian grid. B and D represent respectively in a 2D and 2D-axisymmetric cases, the fully resolved thin film on a fine Cartesian grid. Additionally, in subfigure (A) the definition of the $\hat{\theta}$ is illustrated, which is zero at the curve center. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

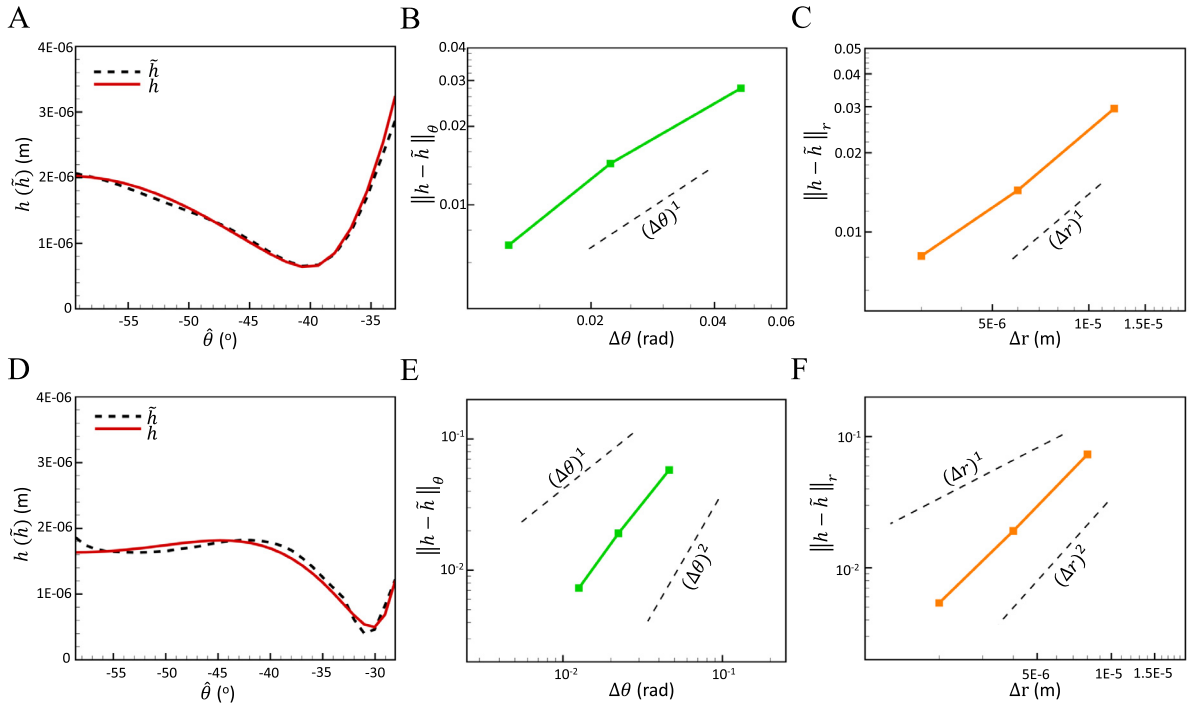


Fig. 6. A and D respectively compare the thickness of subgrid 2D and 2D-axisymmetric thin film, h , captured by the proposed multiscale model, to that of the fully resolved thin film, $\tilde{h}$; B and E present a convergence study in θ -component for the L_2 error of thin-film thickness, for the 2D and 2D-axisymmetric cases, respectively; Similarly, C and F present a convergence study in r -component for the L_2 error of thin-film thickness. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

$$\|h - \tilde{h}\| = \frac{\sqrt{\frac{1}{n} \sum_k (h_k - \tilde{h}_k)^2}}{\tilde{h}_k}. \quad (52)$$

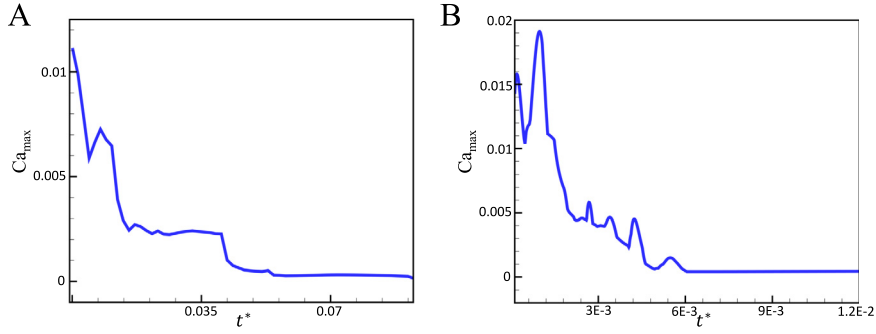


Fig. 7. The spurious currents of the augmented Young-laplace case for the 2D (A) and 2D-axisymmetric (B) scenarios.

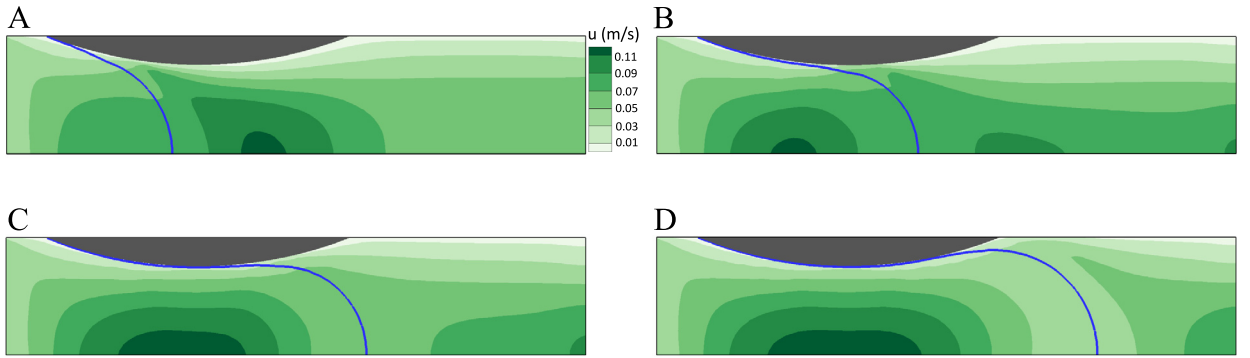


Fig. 8. Snapshots for the drainage process with $Ca = 8E-2$ and $Re = 6E-3$ at A. $t^* = 0.01$; B. $t^* = 0.04$; C. $t^* = 0.08$; and D. $t^* = 0.12$. The color scale represents the magnitude of the horizontal velocity. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Upon this error convergence study, Figs. 6B-C and 6E-F show that the multiscale thin-film lubrication model converges with grid refinement in both azimuthal and radial coordinates by approximately more than first-order.

For a static meniscus on a curved surface, the spurious currents should smear out gradually as a force balance gets established between the surface tension force, the pressure jump across the interface, and the disjoining pressure. Respectively for the 2D and 2D-axisymmetric scenarios, Figs. 7A and 7B show the maximum Capillary number, defined as $Ca_{max} = \mu u_{max} / \sigma$, as a function of dimensionless time t^* , where u_{max} is maximum velocity at each time instance. In Figs. 7A and 7B, the initial value of Ca_{max} correspond to the initially induced spurious currents ($\propto O(u_{max})$) due to the force imbalance caused by the initialization of the interfaces. As the simulation proceeds in time and the system reaches the steady state and a balance between interfacial forces gets established the spurious currents decrease and the maximum Capillary number approaches zero ($Ca_{max} \rightarrow 0$).

4.2. Dynamic thin-films and advancing fluid-fluid interfaces on a curved surface

In the following two subsections, we impose a constant injection flow rate with a corresponding characteristic velocity of U where a non-wetting (or wetting) fluid displaces a wetting (or non-wetting) fluid in a drainage (or imbibition) scenario in a converging-diverging capillary tube. In both scenarios, we consider oil as the non-wetting fluid with a density of $\rho_1 = 878.7 \text{ kg/m}^3$ and viscosity of $\mu_1 = 0.287 \text{ Pa s}$, and consider water as the wetting fluid with a density of $\rho_2 = 1042 \text{ kg/m}^3$ and viscosity of $\mu_2 = 0.001 \text{ Pa s}$. The surface tension between the oil and water is $\sigma = 70 \text{ mN/m}$. The value of the Hamaker constant accounting for the effects of disjoining pressure is $A = -1E - 19$. In both drainage and imbibition scenarios, we investigate the impacts of several factors such as viscosity ratio, Reynolds number, Capillary number, and Hamaker constant on the thin-film profile.

The geometry of the domain and the computational resolution considered in the drainage and imbibition scenarios are identical. The size of the computational domain is $[480\mu\text{m} \times 100\mu\text{m}]$, the radius of the curved surface is $R = 340.2 \mu\text{m}$, and the angular span of the curved surface ranges from -21° to 21° (i.e., 4.71 to 5.07 radians). By applying the multiscale model, we solve the Navier-Stokes-Level-Set on a Cartesian grid with a resolution of $[n_z, n_r] = [120, 25]$, with a corresponding Cartesian coordinate grid size of $\Delta x = \Delta y = 4 \mu\text{m}$. In order to capture the evolution of the subgrid thin-film profile on the Cylindrical coordinate, we solve Eq. (15) with a Cylindrical grid resolution of $\Delta\theta \approx 0.01 \text{ rad}$ and $\Delta r = 10 \mu\text{m}$.

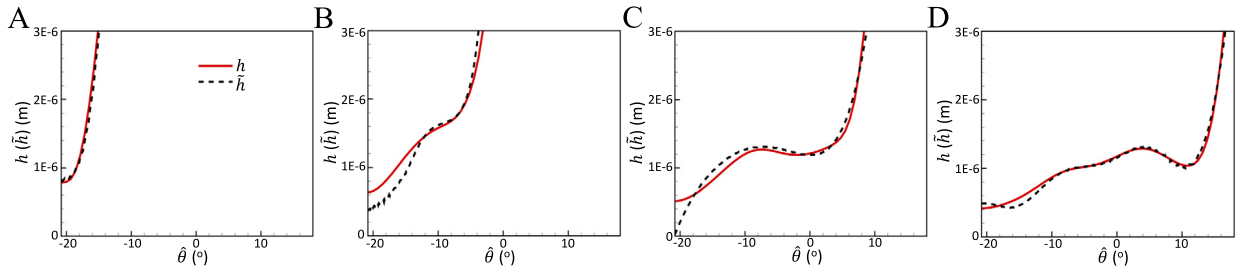


Fig. 9. The thin-film profiles in the drainage process with $Ca = 8E-2$ and $Re = 6E-3$ at different non-dimensionalized times, corresponding to the sequence of the plots in Fig. 8. A. $t^* = 0.01$; B. $t^* = 0.04$; C. $t^* = 0.08$; and D. $t^* = 0.12$. The red line represents the subgrid thin film solved by our multiscale model, while the black dashed line shows the fully resolved thin film on a much higher resolution, $[n_z, n_\zeta] = [960, 200]$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

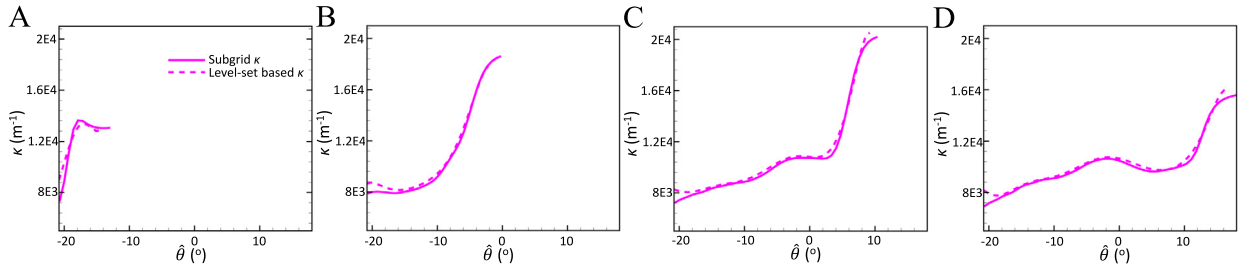


Fig. 10. The profiles of the curvature in the drainage process with $Ca = 8E-2$ and $Re = 6E-3$ at different non-dimensionalized times, corresponding to the sequence of the plots in Fig. 8. A. $t^* = 0.01$; B. $t^* = 0.04$; C. $t^* = 0.08$; and D. $t^* = 0.12$. The solid line represents curvature calculated solved from Eq. (18), while the dashed line shows the curvature interpolated by level-set. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

4.2.1. Advancing non-wetting fluid displacing a wetting fluid - drainage

The displacement of a wetting fluid by a non-wetting fluid falls into a drainage scenario. Fig. 8 shows the snapshots of instantaneous phase interface for a drainage process occurring between oil and water with a corresponding injection velocity of $U = 0.02$ m/s, at different dimensionless time $t^* = t/(2R/U)$ instances. Using the properties of the injected non-wetting fluid (oil) mentioned above, the Capillary number and Reynolds number in the drainage scenario respectively become $Ca = \mu_1 U / \sigma = 8E-2$ and $Re = \rho_1 U R / \mu_1 = 6E-3$. Fig. 8 illustrates that the meniscus advances smoothly on a precursor thin film and no discontinuous velocity jumps occur in the domain, which demonstrates the accuracy and capability of the present method on modelling this scenario.

In Fig. 9, we plot the thin-film profile corresponding to the drainage process shown in Fig. 8. Similar to the study of a static meniscus in Section 4.1, we compare the subgrid thin-film profile solved by our multiscale model (h) with the fully resolved thin-film profile ($\tilde{h}$) directly resolved on a much higher grid resolution, as a verification of our multiscale model. These two solutions reach an agreement on the evolution of the thin film. At early times (Fig. 9A), the thickness of the thin film behind the film front is low and approximately constant. Then, the viscous shear forces lead to the deformation of the thin film and transitioning to a relatively thicker film behind the film front (Fig. 9B). As soon as the thin film's front passes through the middle of the curved surface (i.e., $\hat{\theta} = 0^\circ$), the thin-film profile bends near the center of the curved surface (Fig. 9C). This bend advances along with the front of the thin film till it almost covers the entire curved surface (Fig. 9D).

Fig. 10 compares the profile of the interface curvature (κ) computed using the analytical model expressed by Eq. (18) versus the curvature calculated using the value of the level-set function and Eq. (7). As the subgrid thin film is below the grid resolution in the multiscale model, we interpolate the level-set function on the thin film's surface from the nearby cells. The comparison between the solid and dashed curves in Fig. 10 indicates that the two ways of computing the interface curvature provide relatively similar values, demonstrating that our subgrid thin-film model accurately resolves the interface curvature. More importantly, the curvature profile corresponds to the geometry of the phase interface close to the curved solid. The curvature has a low value when the film is approximately parallel to the curved solid surface. Away from the curved solid surface, the curvature increases to its value at the interface front. Comparing Fig. 10 to Fig. 9 at each time frame, we find that the evolution of the curvature profile follows a similar trend as the evolution of the thin-film profile, which indicates that the surface tension forces play an important role in setting the dynamics of the advancing interface.

To further evaluate the effects of the capillary forces, viscous forces, and inertial forces on the drainage process, we present the thin-film profiles with different Capillary numbers and Reynolds numbers in Fig. 11, where we scale Capillary number and Reynolds number with the values used in Figs. 8 and 9. The red line and symbols in Fig. 11 represent the case shown in Figs. 8 and 9, where $Ca = 8E-2$ and $Re = 6E-3$. We first keep the Reynolds number constant but vary the Capillary number by changing the surface tension (σ). By comparing the differently colored solid lines in Fig. 11, we find

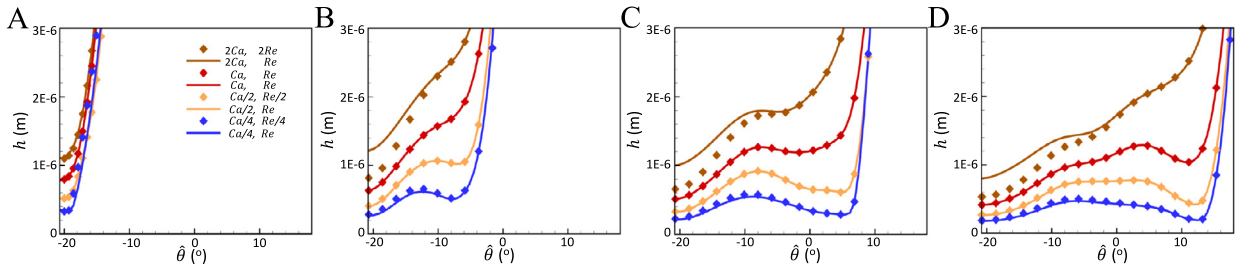


Fig. 11. The comparison of the thin-film profile in the drainage process with different Capillary and Reynolds numbers which are scaled by the values of Ca and Re used in the case shown in Figs. 8 and 9. A. $t^* = 0.01$; B. $t^* = 0.04$; C. $t^* = 0.08$; and D. $t^* = 0.12$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

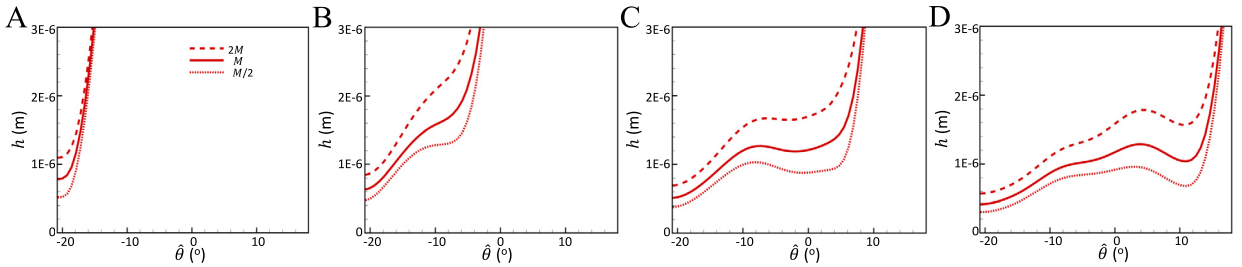


Fig. 12. The comparison of the thin-film profile in the drainage process with same Capillary number and Reynolds number but different viscosity ratio scaled by the viscosity ratio used in the case shown in Figs. 8 and 9 at A. $t^* = 0.01$; B. $t^* = 0.04$; C. $t^* = 0.08$; and D. $t^* = 0.12$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

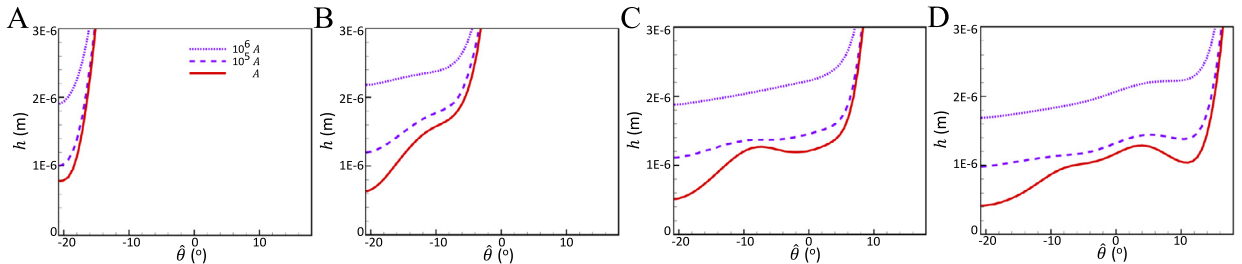


Fig. 13. The comparison of the thin-film profile in the drainage process with different Hamaker constant, at A. $t^* = 0.01$; B. $t^* = 0.04$; C. $t^* = 0.08$; and D. $t^* = 0.12$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

that the thin-film thickness decreases with decreasing the Capillary number. Moreover, the front of the thin film moves faster at lower Capillary numbers. Afterwards, we keep the Capillary number constant but change the Reynolds number by modifying the injection velocity, U . By comparing the solid curves and the discrete symbols with the same color in Fig. 11, we do not observe a significant difference caused by varying the Reynolds number. This could be due to the fact that the value of the Reynolds number considered in this study is relatively low (i.e., $Re \ll 1$).

We also study the effect of the viscosity ratio of the displaced wetting fluid to the injected non-wetting fluid on the evolution of the thin film in the drainage process. In Fig. 12, we fix Reynolds number and Capillary number to be $6E-3$ and $8E-2$, respectively, but change the viscosity ratio by varying the viscosity of the wetting fluid (μ_2). All three used viscosity ratios are scaled with the value used in the case shown in Figs. 8 and 9, which is $M = \mu_2/\mu_1 = 0.001/0.287 = 1/287$. The comparison shown in Fig. 12 indicates that decreasing the viscosity ratio reduces the thin-film thickness. Moreover, the advancing speed of the thin film's front is less at higher viscosity ratios. This could be due to the viscous forces exerted by the more viscous displaced wetting fluid.

The last factor we evaluate is the Hamaker constant. In the case shown in Figs. 8 and 9, we set the Hamaker constant to be $A = -1E - 19$ J. In Fig. 13, we increase the magnitude of A by several orders to investigate the effect of Hamaker constant on the thin-film profile. We find that the thin film becomes thicker and deforms less at larger values of the Hamaker constant.

4.2.2. Advancing wetting fluid on a precursor thin-film that wets the solid - imbibition

In this section, we continue to investigate the dynamics of the augmented Young-Laplace problem by considering an imbibition scenario where a wetting fluid (water) displaces a non-wetting fluid (oil) in the presence of a constant injection

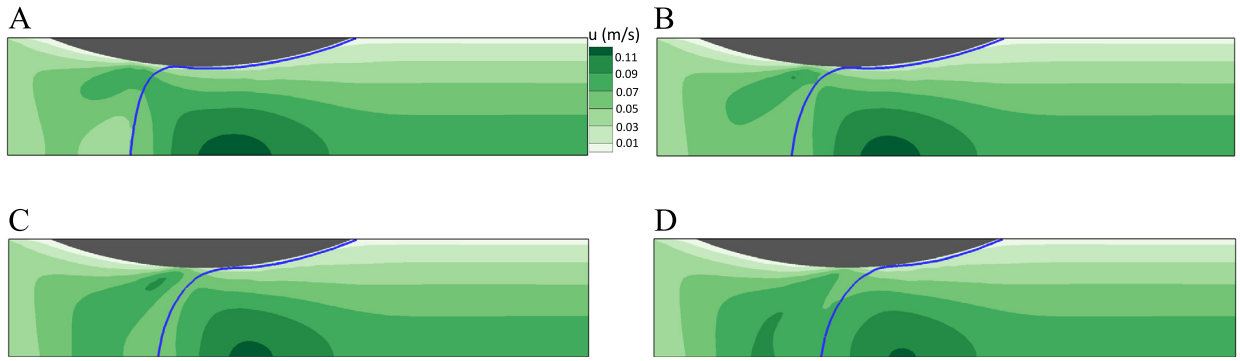


Fig. 14. Snapshots for the imbibition process with $Ca = 1E-4$ and $Re = 2$ at A. $t^* = 0.006$; B. $t^* = 0.012$; C. $t^* = 0.018$; and D. $t^* = 0.024$. The geometry of the computational domain is same as Fig. 8. The color scale represents the velocity. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

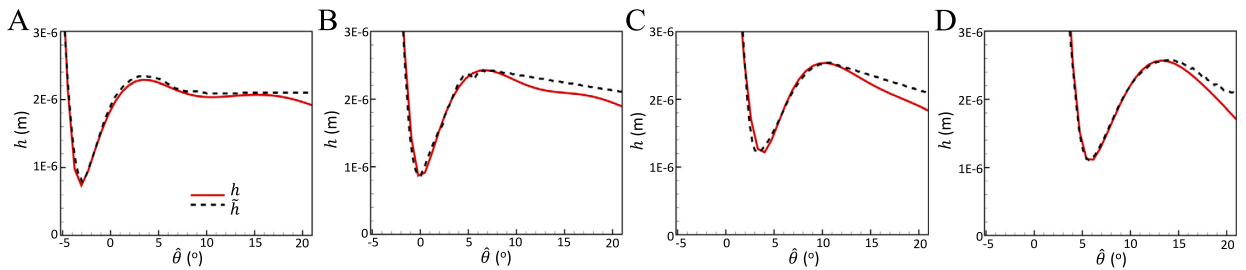


Fig. 15. The thin-film profiles in the imbibition process with $Ca = 1E-4$ and $Re = 2$ at different non-dimensionalized times, corresponding to the sequence of the plots in Fig. 14. A. $t^* = 0.006$; B. $t^* = 0.012$; C. $t^* = 0.018$; and D. $t^* = 0.024$. The red line represents the subgrid thin film solved by our multiscale model, while the black dashed line shows the fully resolved thin-film on a much higher resolution, $[n_z, n_\zeta] = [960, 200]$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

rate. Fig. 14 presents the snapshots of the simulation for an imbibition process with an injection velocity of $U = 0.02$ m/s. Because the injected fluid is water instead of oil in the imbibition process, the Reynolds number and the Capillary number are different from the ones of the drainage process, although the fluid properties and the domain geometry are exactly the same. Using the properties of water, we compute the Capillary number and Reynolds number for the imbibition process shown in Fig. 14 respectively as $Ca = \mu_2 U / \sigma = 1E-4$ and $Re = \rho_2 U R / \mu_2 = 2$.

As mentioned by Abu-Al-Saud et al. [67], due to the high surface tension forces, the numerical simulation at low Capillary number is very sensitive to the accuracy of the curvature computation near the phase interface. Even tiny numerical errors can be amplified by the spurious currents ultimately leading to large numerical perturbations in the flow field. Fig. 14 illustrates the velocity field in the imbibition process and indicates that our model does not cause any sharp velocity jumps even around the interface, which demonstrates our accurate computation of the curvature during advection and reinitialization of the interface at even a low Capillary number.

Fig. 15 compares the evolution of the thin-film profile computed by our multiscale model (h) with the fully resolved thin-film profile ($\tilde{h}$) directly resolved on a much higher grid resolution. Applying the immersed boundary method described in Appendix A, viscous bending is observed where the thin-film profiles are highly skewed, without any externally imposed contact angle boundary condition. The thin film at the viscous bend becomes thinner as the left end of thin film passes through the center of the curved surface (i.e., $\hat{\theta} < 0^\circ$). Comparing the red solid and black dashed curves in Fig. 15, we see that the thin-film solution of the multiscale model agrees well with the fully resolved one, although some deviation exists at the rightmost location where the solid curve intersects with the upper domain flat solid boundary.

In Fig. 16, we plot the profiles of the curvature either calculated from the subgrid analytical model (Eq. (18)) or based on the level-set function using Eq. (7). The comparison between Fig. 15 and Fig. 16 suggests that the maximum value of curvature arises at the viscous bend, as a result of the highly skewed interface. Moreover, by comparing the solid and dashed curves in Fig. 16, we find that the curvature profiles obtained by either of the two approaches (i.e., Eq. (18) or Eq. (7)) agree well with each other over almost the thin-film profile, except at the left boundary where we define the boundary condition of curvature by averaging the level-set curvature with the thin-film curvature, as described in Section 3.2.2. Additionally, the evolution of the curvature profile is consistent with the evolution of the thin-film profile, indicating that the imbibition process, similar to the drainage process introduced in Section 4.2.1, is dominated by the surface tension forces.

Similar to the drainage scenario, we investigate the role of the viscous forces, inertial forces, and capillary forces in the imbibition process by comparing the thin-film profiles with different Reynolds number and Capillary number in Fig. 17. Comparing the solid curves with the discrete symbols having the same color, similar to the drainage scenario, we find that

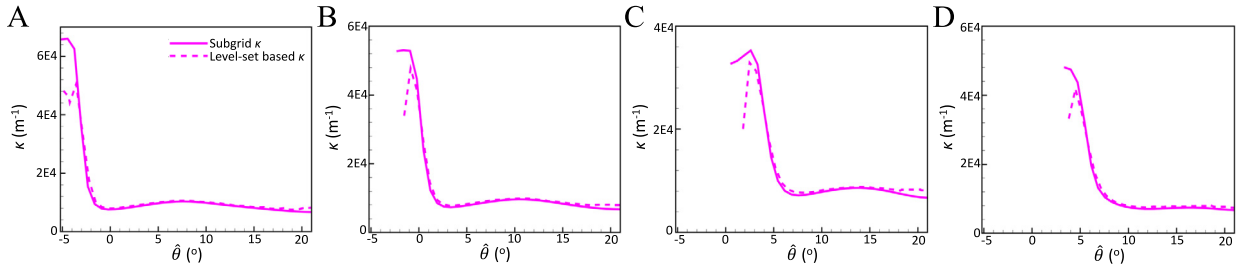


Fig. 16. The profiles of the curvature in the imbibition process with $Ca = 1E-4$ and $Re = 2$ at different non-dimensionalized times, corresponding to the sequence of the plots in Fig. 14. A. $t^* = 0.006$; B. $t^* = 0.012$; C. $t^* = 0.018$; and D. $t^* = 0.024$. The solid line represents curvature calculated solved from Eq. (18), while the dashed line shows the curvature interpolated by level-set. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

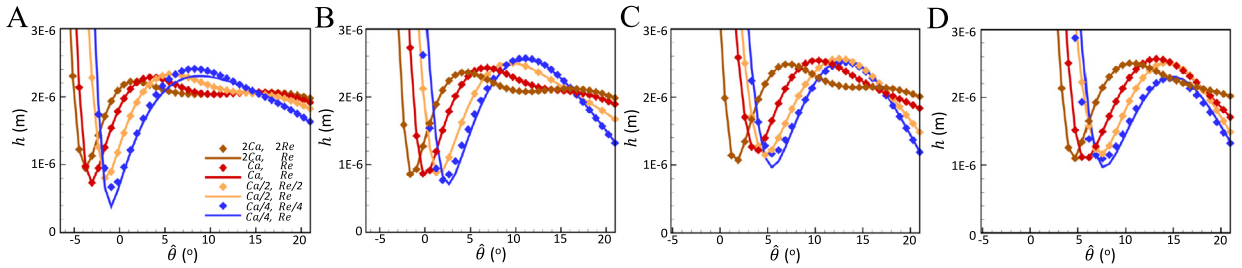


Fig. 17. The comparison of the thin-film profile in the imbibition process with different Capillary and Reynolds numbers which are scaled with the values of Ca and Re used in the case shown in Figs. 14 and 15. A. $t^* = 0.006$; B. $t^* = 0.012$; C. $t^* = 0.018$; and D. $t^* = 0.024$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

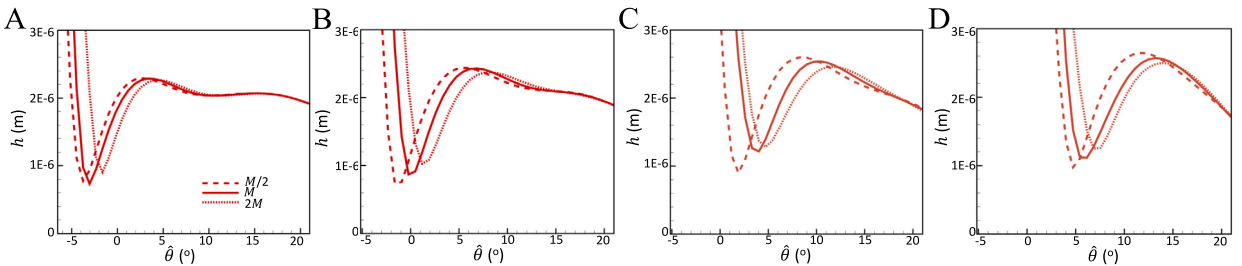


Fig. 18. The comparison of the thin-film profile in the imbibition process with same Capillary number and Reynolds number but different viscosity ratio scaled by the viscosity ratio, M , used in the case shown in Figs. 14 and 15 at A. $t^* = 0.006$; B. $t^* = 0.012$; C. $t^* = 0.018$; and D. $t^* = 0.024$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

the Reynolds number does not affect the thin-film profile significantly, although the Reynolds number is higher than that in the drainage process. However, the Capillary number plays a more important role. The comparison of the solid curves with different colors indicates that the advancing speed of the interface is higher at lower Capillary numbers and the thin-film at the viscous bend is approximately thinner at lower Capillary numbers.

The viscosity ratio of the injected wetting fluid (water) to the displaced non-wetting fluid (oil) also affects the advancing speed of the interface and the thin-film thickness at the viscous bend. Fig. 18 indicates that a lower viscosity ratio leads to a lower advancing speed and a thinner thin film at the viscous bend, as a result of the higher viscous forces exerted from the displaced non-wetting fluid.

The last factor we investigate, which may potentially affect the imbibition process, is the disjoining pressure. Fig. 19 compares the thin-film profiles with different Hamaker constants, suggesting that the effect of the disjoining pressure is not obvious until the magnitude of Hamaker constant gets increased significantly (*i.e.*, five orders of magnitude), beyond which the thin film at the viscous bend becomes thinner and the whole profile becomes smoother with the continuously increasing magnitude of Hamaker constant.

5. Conclusion

In this study, we presented a multiscale continuum model coupling a Navier-Stokes solver on Cartesian coordinate with a topology-preserving level-set method for the numerical simulation of the immiscible two-phase pore-scale flows at low

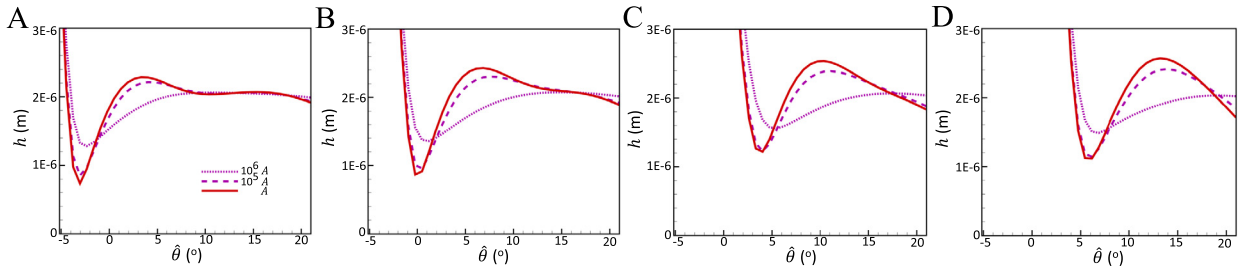


Fig. 19. The comparison of the thin-film profile in the imbibition process with different Hamaker constants, at A. $t^* = 0.006$; B. $t^* = 0.012$; C. $t^* = 0.018$; and D. $t^* = 0.024$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Capillary numbers. In order to capture the evolution of the thin-film on a curved solid surface and couple its effects with the flow dynamics, we presented a lubrication approximation on a cylindrical coordinate.

The proposed model was validated through the simulation of both static and advancing fluid-fluid interfaces in the capillary tubes in the presence of curved solid surfaces. The great agreement between the subgrid thin-film profiles computed by the proposed multiscale model and the fully resolved thin-film profiles demonstrated the accuracy of the proposed model. Even at $Ca < 10^{-4}$, the proposed model does not lead to sharp jumps in the velocity field. The proposed model also predicts the viscous bending in an imbibition process on a curved surface, which has previously been captured in an imbibition process on flat walls [84], without any externally imposed contact angle boundary conditions. Additionally, we applied the proposed multiscale model to investigate the roles of several factors such as Capillary number, Reynolds number, viscosity ratio, and disjoining pressure on the thin-film profile. The simulation results indicate that compared to the inertial forces, the capillary forces play a more significant role in the thin-film evolution at the pore-scale. Besides the capillary forces, the thin film can also be affected by the viscous forces and the disjoining pressure.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Funding for this work was provided by the Reservoir Simulation Industrial Affiliates Program at Stanford University (SUPRI-B).

Appendix A. Immersed boundary approach

The solid-fluid interface at arbitrary time does not coincide with the underlying Cartesian grid. To enforce the desired boundary condition as accurately as possible, we have to first numerically identify the location of the interface between the solid body and the fluid. We use finite difference method to solve the Navier-Stokes equations (Eqs. (2)-(3)) on a staggered grid, where the pressure p exists at the grid centers and the velocity fields exist at the appropriate grid walls, as illustrated in Fig. 20. At each time step, we build a level-set function, ϕ , whose zero level represents the solid-fluid interface, based on the topological parameters of the particles. We consider a grid as solid grid if $\phi < 0$ at the grid center (e.g., the grids marked by red unfilled circle in Fig. 20A). The pressure of the solid grids is not computed explicitly from the Poisson equation, Eq. (28), but the pressure boundary condition Eq. (30) is implicit into the RHS of the Poisson equation through the velocity boundary conditions, $\mathbf{f}_{IB}^{n+1}$ and $\mathbf{b}^{n+1}$. Eq. (25) relates the velocity field to the pressure field, its discretization in u-component can be,

$$u_{i+1/2,j}^{n+1} = \hat{u}_{i+1/2,j} - \frac{\Delta t}{\rho(\mathbf{x})} \frac{p_{i+1,j}^{n+1} - p_{i,j}^{n+1}}{\Delta x}, \quad (\text{A.1})$$

where $(i + 1/2, j)$ represents a grid wall of (i, j) . To make the velocity boundary consistent with the pressure boundary, we consider $(i + 1/2, j)$ as a boundary point (e.g., the grids marked by bigger blue unfilled circle in Fig. 20A) and solve $u_{i+1/2,j}^{n+1}$ out from Eq. (27) instead of Eq. (A.1) if either grid $(i + 1, j)$ or (i, j) is a solid grid. Similarly, in v-component, $(i, j + 1/2)$ is considered to be a boundary point if either grid $(i, j + 1)$ or (i, j) is a solid grid.

In the frame of Direct forcing method [55] adopted in the present work, we consider the velocity boundary condition $\mathbf{f}_{IB}^{n+1}$ and $\mathbf{b}^{n+1}$ directly in the context of the discretized equations. Here, we illustrate the details of our strategy used to enforce the u-component velocity boundary condition, it has the same form for v-component. For the boundary points at first, e.g., point "O" in Fig. 20(A), the u-component of $\mathbf{b}^{n+1}$ is computed using a 2nd order linear interpolation method,

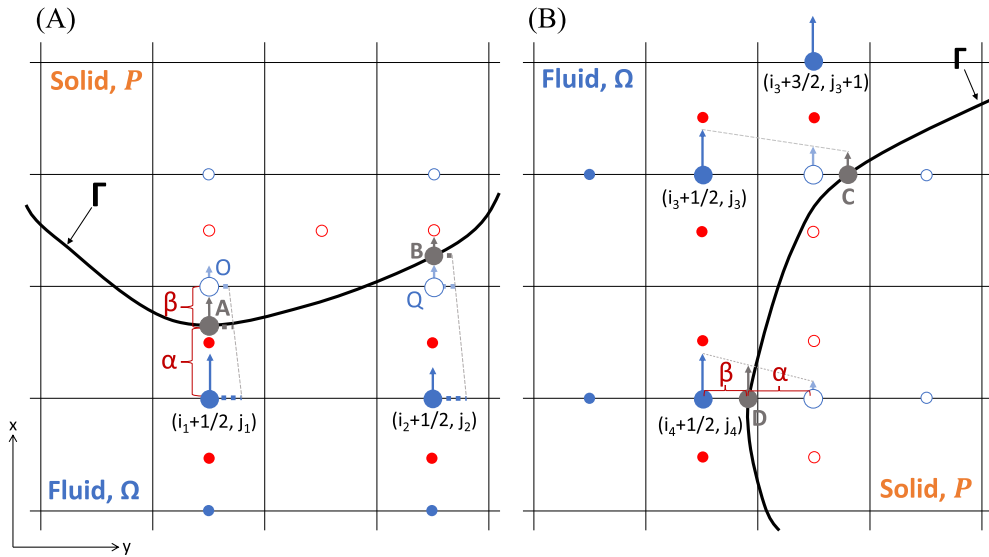


Fig. 20. Illustration of our immersed boundary method. (A) represent the application of the immersed boundary method on the x-component of the discretization in the left-hand side of Eq. (A.3), while (B) represents the application on the y-component. The black curve represents the solid-fluid interface, Γ . Red circles represent the pressure field and the blue circles represent the velocity field. The filled circles are the grids or grid walls in the fluid domain Ω , while the unfilled small circles are the grids or grid walls in the solid domain P . The bigger unfilled blue circles represent the interfacial points where we enforce the boundary condition into. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

$$\alpha(b_u)_O^{n+1} + \beta u_{i_1+1/2, j_1}^n = u_A^n, \quad (\text{A.2})$$

where α and β are constants depending on the relative position between points A, O and $x_{i_1+1/2}$.

Enforcing the boundary condition to the grids relevant to the interfacial points is more complex. How we define “relevant to” is based on the discretization of Eq. (22), as

$$C_{x-r}\hat{u}_{i+3/2, j} + C_{x-l}\hat{u}_{i-1/2, j} + (1 - C_{x-r} - C_{x-l} - C_{y-r} - C_{y-l})\hat{u}_{i+1/2, j} + C_{y-r}\hat{u}_{i+1/2, j+1} + C_{y-l}\hat{u}_{i+1/2, j-1} + (\mathbf{f}_{IB})_{i+1/2, j}^{n+1} = RHS_{i+1/2, j}^n, \quad (\text{A.3})$$

where C_{x-r} , C_{x-l} , C_{y-r} and C_{y-l} are the coefficients including the discrete properties, and the temporal and spatial sizes. Eq. (A.3) indicates that the solution of $\hat{u}_{i+1/2, j}$ is based on the velocity on five relevant points, including $(i+1/2, j)$ itself and two adjacent grids on each x or y direction. We consider a grid to be “relevant to” the interfacial points if any of these four adjacent grids is boundary grid.

Fig. 20(A) and (B) illustrate the scheme by which we enforce $(\mathbf{f}_{IB})_{i+1/2, j}^{n+1}$ into x and y components of the left hand side of Eq. (A.3), respectively. In x component, we treat the body force as a Dirichlet boundary condition. For an example, at grid $(i_1+1/2, j_1)$ in Fig. 20(A), we assume the body force enforces a boundary condition, Eq. (24), into the term $\hat{u}_{i_1+3/2, j_1}$ (at point “O”), so that

$$C_{x-r}\hat{u}_{i_1+3/2, j_1} + (\mathbf{f}_{IB})_{i_1+1/2, j_1}^{n+1} = \mathbf{b}_{i_1+3/2, j_1}^{n+1} + \Delta t \frac{1}{\rho(\mathbf{x})} \left(\frac{\partial p}{\partial x} \right)_{i_1+1/2, j_1}^n, \quad (\text{A.4})$$

where we assume gradient of pressure reduces to zero across the solid-fluid interface, hence $(\frac{\partial p}{\partial x})_{i_1+1/2, j_1}^n = 0$. Giving Eq. (A.4) to (A.3), we obtain the final equation of $\hat{u}_{i_1+1/2, j_1}$ as follows,

$$C_{x-l}\hat{u}_{i_1-1/2, j_1} + (1 - C_{x-r} - C_{x-l} - C_{y-r} - C_{y-l})\hat{u}_{i_1+1/2, j_1} + C_{y-r}\hat{u}_{i_1+1/2, j_1+1} + C_{y-l}\hat{u}_{i_1+1/2, j_1-1} = RHS_{i_1+1/2, j_1}^n - C_{x-r}(b_u)_{i_1+3/2, j_1}^{n+1}. \quad (\text{A.5})$$

On the other hand, in y component, we assume the body force creates a Neumann boundary condition. At the grid $(i_4+1/2, j_4)$ in Fig. 20(B), the body force promotes $u_{i_4+1/2, j_4+1}^{n+1}$ (at point “D”) to satisfy the following 2nd order linear interpolation,

$$\alpha u_{i_4+1/2, j_4}^{n+1} + \beta u_{i_4+1/2, j_4+1}^{n+1} = u_D^{n+1}. \quad (\text{A.6})$$

Combining Eqs (A.6) with Eq. (24), one can obtain a Neumann boundary condition, as

$$\hat{u}_{i_4+1/2, j_4+1} = -\frac{\alpha}{\beta}\hat{u}_{i_4+1/2, j_4} + \frac{1}{\beta}u_D^{n+1} + \frac{1}{\beta}\Delta t \frac{1}{\rho(\mathbf{x})} \left(\frac{\partial p}{\partial x} \right)_{i_4+1/2, j_4}^n + \Delta t \frac{1}{\rho(\mathbf{x})} \left(\frac{\partial p}{\partial x} \right)_{i_4+1/2, j_4+1}^n, \quad (\text{A.7})$$

where we also assume the pressure gradient reduces to zero across the interface, so that

$$\left(\frac{\partial p}{\partial x}\right)_{i_4+1/2, j_4}^n = \left(\frac{\partial p}{\partial x}\right)_{i_4+1/2, j_4+1}^n. \quad (\text{A.8})$$

Giving Eqs. (A.6)–(A.8) to (A.3), we obtain the final equation of $\hat{u}_{i_4+1/2, j_4}$,

$$\begin{aligned} C_{x-r}\hat{u}_{i_4+3/2, j_4} + C_{x-l}\hat{u}_{i_4-1/2, j_4} + (1 - C_{x-r} - C_{x-l} - C_{y-r}(1 + \frac{\alpha}{\beta}) - C_{y-l})\hat{u}_{i_4+1/2, j_4} + C_{y-l}\hat{u}_{i_4+1/2, j_4-1} \\ = R H S_{i_4+1/2, j_4}^n - \frac{C_{y-r}}{\beta} u_D^{n+1} - \frac{(1 + \beta)C_{y-r}}{\beta} \Delta t \frac{1}{\rho(\mathbf{x})} \left(\frac{\partial p}{\partial x}\right)_{i_4+1/2, j_4}^n. \end{aligned} \quad (\text{A.9})$$

As introduced above, our approach uses one-dimension linear interpolation in each x or y direction, with which the boundary points are particularly the grids closest to the solid body. Unlike other numerical methods (e.g., [59]), our approach allows the boundary points to exist both inside and outside the solid-fluid interface, for examples, the points O and Q in Fig. 20(A). It tends to improve the stability of the computation and is more efficient than other forcing strategies such as feedback forcing [55]. Additionally, we engage the body force as Dirichlet or Neumann into the different component of discretization. This strategy eliminates the potential ambiguity about the direction of interpolation, which is mentioned in [59]. This improvement is illustrated by Fig. 20(B). A boundary condition at $(i_3 + 1/2, j_3 + 1)$ (represented by bigger blue unfilled circle) is needed by the y-component of the discretization of $\hat{u}_{i_3+1/2, j_3}$. According to our interface identification, $(i_3 + 1/2, j_3 + 1)$ is a boundary point, where $(b_u)_{i_3+1/2, j_3+1}^{n+1}$ has been defined by Eq. (A.2) and has been enforced as a Dirichlet boundary condition into the x-component of the discretization of $\hat{u}_{i_3+3/2, j_3+1}$. To avoid the duplicate definition, we engage the body force at $(i_3 + 1/2, j_3 + 1)$ as a Neumann boundary condition, which is same as Eq. (A.6) by replacing i_3 by i_4 , instead of Dirichlet boundary condition into the y-component of the discretization of $\hat{u}_{i_3+1/2, j_3}$.

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